

AERO62520 Coursework-4 Preliminary Design Review

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1 Introduction

Global industries—from logistics and agriculture to disaster response—are confronting acute labor shortages and mounting pressure to improve operational efficiency, creating an urgent need to automate repetitive, hazardous, and resource-intensive tasks. Future automation must go beyond simple point-to-point motion and instead combine autonomous navigation, real-time object recognition, and precise manipulation to operate reliably in unstructured, unpredictable environments. This project responds to that need by targeting tasks that reduce human exposure to risk while increasing throughput and adaptability in dynamic settings.

The objective is to design and develop an autonomous rover capable of independently exploring unknown terrain, identifying target blocks via computer vision, retrieving them with an integrated robotic arm, and delivering loads to designated collection points. Achieving this requires tight coordination among sensing, planning, and actuation subsystems, and the system is intended to simulate real-world industrial collaboration scenarios where robustness and repeatability are essential. Key subsystems include SLAM for localization and mapping, inverse kinematics for accurate arm control, and a mechanically stable chassis sized for expected payloads.

This Preliminary Design Review evaluates whether the proposed architecture—from mechanical feasibility and component selection to software modularization and power budgeting—meets the mission requirements and technical constraints. The review documents 3D CAD models and load evaluations, connection diagrams and energy budgets, and a software deployment plan that partitions perception, planning, and low-level control across processing units. It concludes with a requirements verification matrix and a phased implementation roadmap to guide development toward a validated, manufacturable solution. Specifically, this document includes:

- **System Architecture:** Hardware block diagram and interconnections.
- **Component Analysis:** Description, justification, and purpose of each subsystem.
- **Mechanical Design:** 3D CAD model of the rover, payload sled design files, and discussion of placement trade-offs.
- **Manufacturability and Suitability:** Assessment of material strength, stability, and fabrication feasibility.
- **Power System:** Connection diagram with values and units, power budget, and efficiency considerations.

- **Software Architecture:** Block diagram detailing software modules and their deployment (e.g., NUC or Raspberry Pi).
- **Design Validation:** Preliminary verification against requirements using a verification matrix.
- **Project Plan:** Timeline of milestones, task allocation, and repository links for version-controlled designs.

By addressing trade-offs, manufacturability, and integration challenges at this stage, the PDR ensures that the design is technically sound, scalable, and ready for transition to detailed design and prototyping. This review establishes confidence in the system's ability to fulfill its mission in complex, real-world environments while demonstrating the potential of robotics to automate essential tasks.

2 System

2.1 System Block-Diagram

The hardware connections of the system are shown in Figure 1.

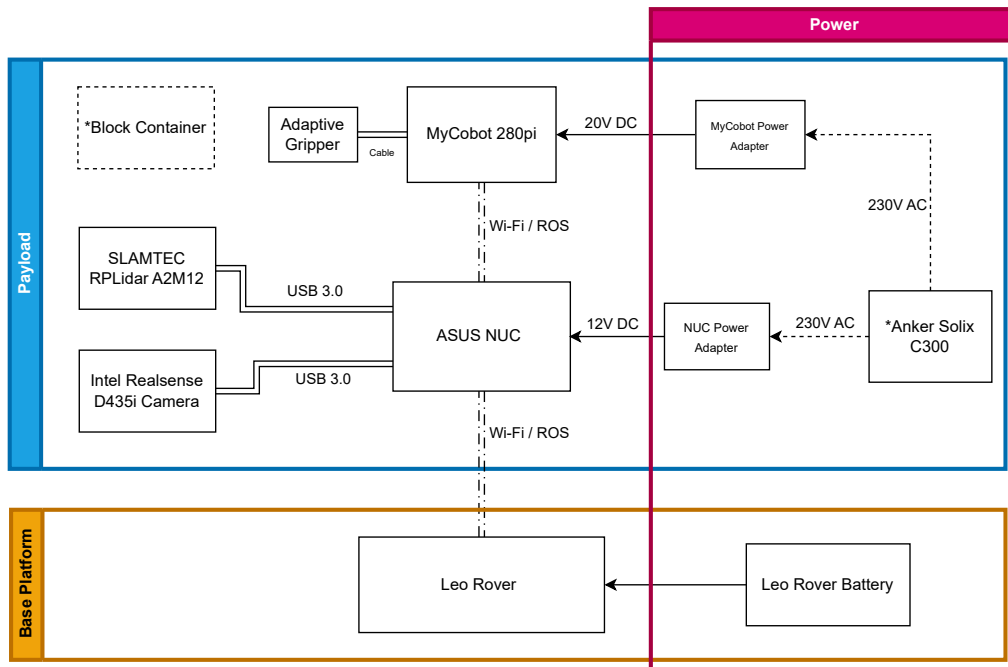


Fig. 1. System Architecture and Power Distribution Block Diagram

Note:

- **Anker Solix C300:** Additional Power Supply. For details, please refer to the 4 section.
- **Block Container:** Mechanical Only

2.2 Component Selection Matrix

Table 1 below presents detailed descriptions and justification for each core component.

Table 1. System Component Specifications and Selection Rationale

Component / Subsystem	Key Specifications	System Function	Justification & Design Rationale
Leo Rover 1.8 (Mobility Platform)	• Payload: 5 kg (Nominal) • Drive: 4WD, Independent Suspension • Core: Raspberry Pi 4B	Provides mobility capabilities to transport the payload payload across the arena and navigate unstructured terrain.	Payload Matching: The 5 kg capacity sufficiently supports the NUC, manipulator, and auxiliary battery (Total ≈ 4.6 kg). Terrain Adaptability: Independent suspension ensures stability when traversing obstacles or cable guards in the lab environment.
ASUS NUC 12 Pro (Computational Core)	• CPU: i7-1260P (12 Cores) • RAM: 16 GB • I/O: USB 3.2 Gen 2	Executes high-load algorithms including SLAM navigation, YOLOv8 object detection, and MoveIt trajectory planning.	Compute Demand: Only x86 architecture with high clock speeds can satisfy the real-time inference requirements (> 30 FPS) for YOLO, which exceeds Raspberry Pi capabilities. Bandwidth: High-speed USB ports ensure simultaneous data ingestion from LiDAR and RGB-D cameras without bottlenecks.
Intel RealSense D435i (Vision Perception)	• Tech: Active Stereo IR • Range: 0.3 m – 3 m • Feature: Global Shutter	Provides RGB streams for object classification and aligned depth clouds for 3D spatial coordinate extraction.	Motion Robustness: The Global Shutter sensor minimizes motion blur during robot movement, ensuring reliable color block detection. Positioning: The depth data allows direct deprojection of 2D pixels to 3D (x,y,z) coordinates essential for inverse kinematics (FR-09).

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Table 1. System Component Specifications and Selection Rationale (Continued)

Component / Subsystem	Key Specifications	System Function	Justification & Design Rationale
SLAMTEC RPLIDAR A2M12 (LIDAR Navigation)	• Range: 12 m • Sample Rate: 16k/sec • Resolution: 0.9°	Delivers 360° environmental scanning data for 2D mapping (SLAM) and obstacle avoidance.	Coverage: The 12 m range fully covers the CW3 arena dimensions. Stability: High sampling rate prevents map drift during rapid rotations, meeting the autonomous navigation requirements.
myCobot 280 Pi (Manipulator)	• DOF: 6-Axis • Payload: 250 g • Repeatability: ± 0.5 mm	Executes end-effector manipulation tasks: approaching, grasping, transporting, and depositing target blocks.	Dexterity: 6-DOF design allows approach from arbitrary angles, adapting to different pickup heights. Capacity: 250 g payload exceeds the maximum block weight of 150 g (PR-04). Lightweight: Low self-mass (850 g) minimizes the impact on the rover's center of mass.
Adaptive Gripper (End Effector)	• Stroke: 20 mm – 75 mm • Interface: LEGO Compatible • Control: Digital I/O	Physically grasps objects of varying sizes securely.	Size Compatibility: The stroke range perfectly covers the 20 mm standard blocks and potential larger obstacles. Simplicity: Simple I/O control reduces software complexity compared to dexterous hands, increasing reliability.
Anker Solix C300 (Additional Power)	• Capacity: 288 Wh • Output: AC 230V , DC 12V • Weight: ≈ 2.3 kg	Provides independent power to the NUC and Manipulator, isolating the load from the rover base.	Reliability: Separation of power ensures that high computational loads do not drain the mobility battery. Integration: AC output allows the use of proprietary adapters for NUC and Arm, simplifying voltage regulation safety compliance.

3 Mechanical Design

3.1 Design Overview

Full Assembly Diagram

Figure 2 illustrates the overall layout, the relative positions of key components, and the spatial relationship between the load pallet and the joint's working space.

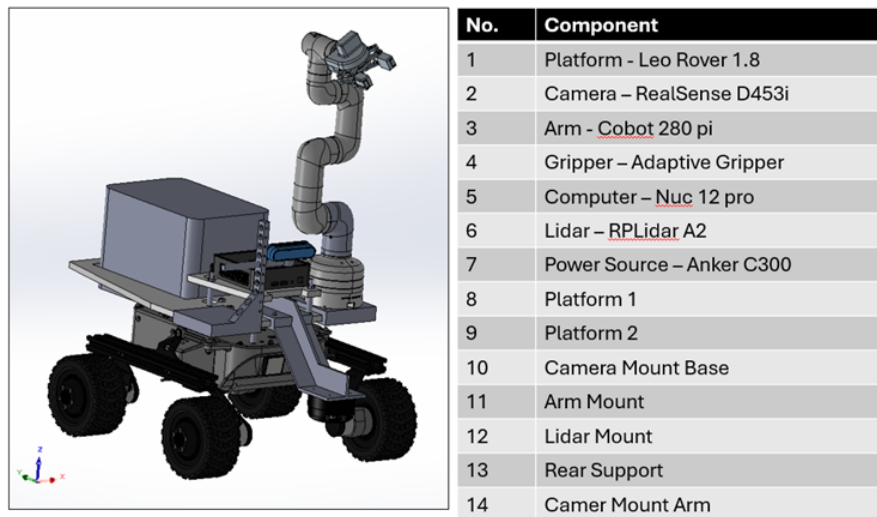


Fig. 2. Overall Assembly (Isometric). Shows the main hardware modules and component list.

Figure 3 clarifies the mounting interfaces, installation heights, and relative positioning of each component, facilitating subsequent design verification and assembly process documentation for the load pallet, camera field of view, and LiDAR field of view.

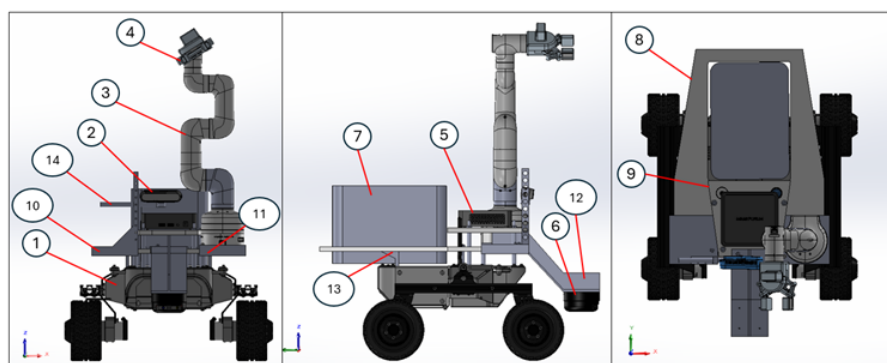


Fig. 3. Three-view drawings with component annotations (Front / Side / Rear). Part numbers correspond to the parts list for assembly and manufacturing instructions.

General Design Table

Table 2 summarizes key decisions in this robotic mechanical design: material and process selection, modular assembly strategies, manufacturability and safety considerations, and mass estimation for the payload pallet. This information supports subsequent detailed assembly drawings and manufacturing documentation.

Table 2. General Features and Design Rationale

General Feature	Description and Rationale
Material Usage	• A combination of laser cut sheets and 3D printed parts have been selected. • Stacked laser cut sheets have been used as platforms and partial enclosures to retain and mount components, inspired by TurtleBot. • Laser cutting used where possible to reduce lead times due to faster process speed. • ABS is the preferred choice of material due to its increased toughness and resistance to brittle fracture; however, its use is only essential for component 14.
Modular Design	• Components mechanically interlock and locate and are either retained (to main body) with a single fastener (or 3D printed pin) or simply enclosed. This aims to save many lab hours by facilitating rapid assembly/disassembly, subsystem debugging, and general group work. • All electrical components are fastened with M4 × 20 screws consolidating fixing type. • The design allows a level of flexibility of mounting locations, with the arm and the camera positions being interchangeable. This extends the lidar with minor modifications. • Allows redesign/repair of single subsystem/mount if required.

Continued on next page...

Table 2 – continued from previous page

General Feature	Description and Rationale
Component Location	• Due to modular design component placement is not 100% fixed and alterations possible with minor modifications. • A general attempt to balance the robot and maintain the centre of mass has been made. • Lidar: Position driven by goal of capturing the 200 mm cubed Coloured Object Plinths which would not have been possible with mounting at Platform 1 height. 200 degrees field of view has been preserved to ensure that the navigation subsystem can collect the required data. This comes at the cost of increasing the minimum distance from the robot to the coloured objects, increasing the stresses on the mounting platforms due to greater arm extension. • Camera: Positioned for forward approach to Coloured object plinths. Central position for optimal viewing of coloured blocks mindful of robotic task space. • Arm: Frontal position to provide adequate grasping range, and off-centre to reduce camera occlusion during grasping and transportation. • Power Source: Mounted centrally on the rear on the robot to maintain stability and act as a counterweight to the other components. • NUC: Mounted in an easily accessible location with clearance for cables and cable management. Unobstructed airflow for thermal regulation.
Future Proofing	• If required, Platform 2 can be exchanged for an alternative that further retains the onboard power source. • The Lidar Mount design provides an interface for additional 3D printed Block storage if required (see component breakdown for details). • Additional 3D printed parts may be added for cable management.
General Robustness	More material than structurally necessary has been used to mitigate potential damage caused by drops, transport, and improper storage.

3.2 Component Breakdown

Camera Mount Assembly - Component 10, 14

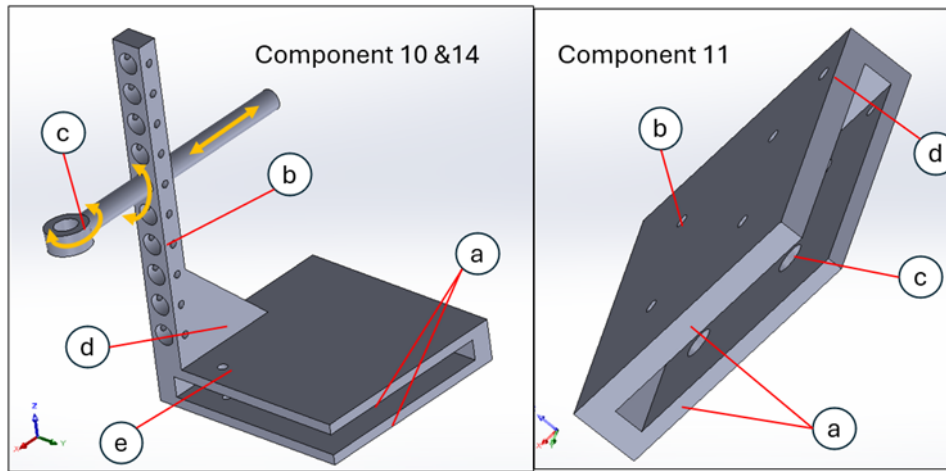


Fig. 4. Camera Mount and Arm Mount Perspective Detail Views. Left: Camera Mount Assembly (Components 10 & 14). Right: Arm Mount (Component 11).

The camera mounting assembly, as shown in Figure 4 and Table 3, is manufactured using 3D printing technology with ABS or PLA material.

Table 3. Key Features and Rationale of the Camera Mount Assembly

Feature	Description and Rationale
a	“Sleeve” design with free running fit. Mechanically locates and interlocks for rapid assembly requiring only a single fastener.
b	Adjustable height camera mount for testing purposes. Can be replaced with static structure when optimal camera position is located. Secured with M2 grub screw into ABS camera mount arm. Additional grub screw location on opposite side for redundancy.
c	Adjustable arm with 3 D.O.F allowing for camera adjustment. M6 interference fit for threaded rod in camera base to tap. Can be replaced with static structure when optimal camera position is located. ABS material necessary.
d	Structural reinforcement at stress concentration point.
e	M4 through hole for fastening to Platform 1.

Arm Mount – Component 11

The arm mount, as shown in Figure 4 and Table 4, is manufactured using 3D printing with ABS or PLA material.

Table 4. Key Features and Rationale of the Arm Mount Design

Feature	Description and Rationale
a	“Sleeve” design with free running fit. Mechanically locates for rapid assembly. Mechanically interlocks requiring only a single fastener.
b	Arm base fastening locations. M4 through holes. Recessed internal cavity for screw heads preventing collision with “sleeve” mechanism.
c	Access holes to enable fastening of Cobot 270 pi arm to Arm Mount.
d	M4 through hole for fastening to Platform 1.

Lidar Mount – Component 12

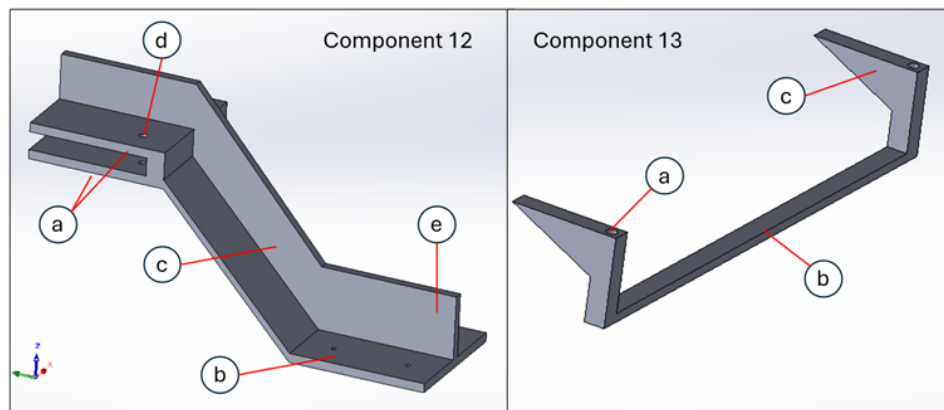


Fig. 5. Lidar Mount and Rear Support Perspective Detail Views. Left: Lidar Mount Assembly (Components 12). Right: Rear Support (Component 13).

The lidar mount, as shown in Figure 5 and Table 5, is manufactured using 3D printing with ABS or PLA material.

Table 5. Key Features and Rationale of the Lidar Mount Design

Feature	Description and Rationale
a	“Sleeve” design with free running fit. Mechanically locates for rapid assembly. Mechanically interlocks requiring only a single fastener.
b	M3 through holes for fastening of Lidar to Mount.
c	Web for increased structural support, stiffness and vibration reduction.
d	M4 through hole for fastening to Platform 1.
e	Extended web provides an interface for a possible block carrying receptacle serving as a mounting point for a vertically orientated “sleeve”.

Rear Support – Component 13

The rear support, as shown in Figure 5 and Table 6, is manufactured using 3D printing with ABS or PLA material.

Table 6. Key Features and Rationale of the Rear Support

Feature	Description and Rationale
a	M4 interference fit hole for mounting screw to tap.
b	Restraining “strap” to mechanically fix support bottom and increase load distribution area.
c	Reinforcement of stress concentration points and increased load distribution area to increase robustness to dynamic effects.

Platform 1 - Component 8

As shown in Figure 6 and Table 7, it is manufactured from a 10-millimetre-thick plastic sheet (acrylic) using laser cutting technology.

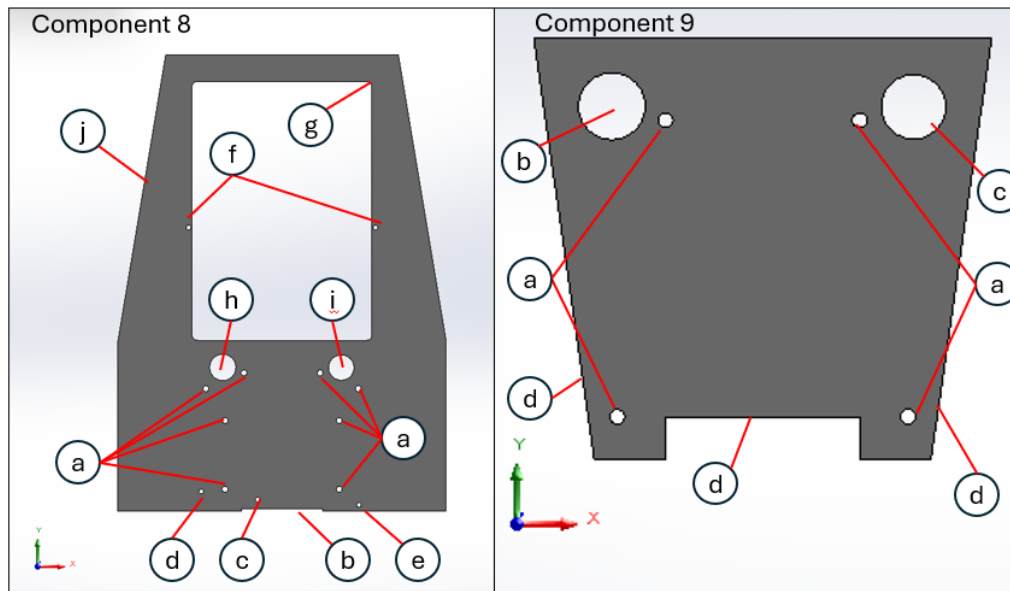


Fig. 6. Platform 1 and Platform 2 Perspective Detail Views. Left: Platform 1 (Components 8). Right: Platform 2 (Component 9).

Table 7. Key Features and Rationale of Platform 1 Design

Feature	Description and Rationale
a	M5 through mounting holes for M5 threaded rod to mount to rover chassis.
b	Locating slot for easy Lidar Mount alignment.
c	M4 through hole for Lidar Mount.
d	M4 through hole for Camera Mount Assembly.
e	M4 through hole for Arm Mount.
f	M4 through holes for Rear Support.
g	Slot to retain onboard power source. Large clearance (approx. 5 mm) given due to unverified dimensions. Rounded internal corners to reduce peak stress at stress concentration points.
h	Antenna clearance hole.
i	Rover External Port clearance hole.
j	Minimal lightweighting whilst maintaining part stiffness and strength.

Platform 2 - Component 9

As shown in Figure 6 and Table 8, it is manufactured from a 10-millimetre-thick plastic sheet (acrylic) using laser cutting technology.

Table 8. Key Features and Rationale of Platform 2 Design

Feature	Description and Rationale
a	M5 through mounting holes for M5 threaded rod to mount to rover chassis.
b	Antenna clearance hole.
c	Rover External Port clearance hole.
d	Clearance for Lidar Mount fixing, Arm task space, Camera adjustment.

Fixings

Table 9 details the complete bill of materials for the mechanical fasteners used in the assembly, including their specific applications. M4 fixings have been used where not constrained by existing components.

Table 9. Summary of Required Fixings and Fasteners

Qty.	Fixing	Notes
4	M5 × 100 threaded rod	Rover chassis to Platform 1/2
4	M5 × 50 threaded rod	Rover chassis to Platform 1
5	M4 × 20 screws	Arm, Lidar, Camera Mounts to Platform 1. Rear support to Platform 1.
4	M4 × 10 screws	Arm to Mount.
4	M3 × 10 screws	Lidar to Mount.
1	M6 × 20 threaded rod	Camera to Mount.
12	M4 × 30 (nylon) spacers	For Platform separation on M5 threaded rod.

Lightweighting

Additional lightweighting was considered however due to the criticality (and lack of spares) of each manufactured component the risk of mechanical failure was unacceptable.

Mass of Sled

The total mass of the payload sled was calculated to be 1.275kg assuming acrylic for lazer-cut component and ABS for 3D prints. A detailed mass breakdown of individual components is presented in Table 10. Platform 1 accounts for the majority of the structural mass due to its role as the primary chassis.

Table 10. Mass Breakdown of Structural Components (Sled Assembly)

Structural Component	Mass (g)
Platform 1	708
Platform 2	188.25
Lidar Mount	127.5
Arm Mount	117
Camera Assembly	126
Rear Support	9
Total	1275

3.3 Further Analysis

Structural analysis was deemed unnecessary given the scope of the project, relatively static loading, and material thickness.

FEA of Platform 1 might be informative for locating stress concentration points and guiding hole positioning, given that parts of this nature are often susceptible to brittle fracture.

The orientation of layers in the 3d prints is critical in maintaining the strength of the parts and will have to be printed off plane.

4 Electrical Design

4.1 Mission Scope and Duration

The onboard power system shall be sized to support a complete operational cycle of the rover: autonomous navigation and arena scanning using onboard cameras, obstacle detection and wall-margin avoidance using the LIDAR, manipulation of a target block (including approach, grasp, transport back to the base, and placement into the designated bin), and all associated sensing, computation, telemetry, and safety functions. The power supply must reliably sustain at least one full cycle of this mission loop and provide continuous untethered operation for a minimum of 20 minutes, as specified by stakeholder requirements.

4.2 Theoretical Power Draw on Component Power Draw Specifications and Mission Scope

Component power consumption specifications and theoretical power consumption for the task scope are shown in Table 11.

Table 11. Component Power Consumption Estimation

Component	DC Volt (V)		Current (A)		Wattage (W)		Dur. (Hr)	Energy (Wh)	
	Min	Max	Min	Max	Min	Max		Min	Max
Asus NUC (Compute)	12	20	Unsp.	10	Unsp.	120	0.33	-	39.96
myCobot 280pi	12	12	5	5	60	60	0.33	19.98	19.98
myCobot 280pi Adaptive Gripper	Unsp.	5	Unsp.	Unsp.	-	-	0.33	-	-
Intel RealSense D435i	4.75	5.25	Unsp.	0.700	Unsp.	3.66	0.33	-	1.21
RPLiDAR A2M12	5	5	0.45	0.60	2.25	3.00	0.33	0.74	0.99
Leo Rover 1.8	11.1	Unsp.	Unsp.	8	Unsp.	Unsp.	0.33	-	-

Note: Unsp. = Unspecified. Wattage calculated as Current $\times$ Voltage. Data derived from component specifications.

4.3 Practical Design Considerations and Calculations

Practical design considerations and calculations are shown in Table 12.

Table 12. System Power Budget and Engineering Rationale

Item(s)	Parameter / Metric	Value / Calculation	Engineering Design Choice Rationale
Mission Scope	Min. Adjusted Duration	35 min (0.583 h)	The duration includes a conservative 75% operational headroom applied to the estimated runtime. This safety measure ensures battery sufficiency and absorbs contingencies like task retries and real-world battery degradation.
Rover Base	Power Requirement	None	The base operates independently on its integrated battery for a minimum of 1 hour; no external power source is required.
NUC (ASUS)	Max Power Draw (Rated)	120 W	Assumed maximum capacity to sustain continuous, intensive computational loads (SLAM, YOLOv8/v11, vision algorithms).
	Effective Power Draw	114 W (120 W $\times$ 0.95)	Applies a conservative 95% efficiency factor to account for thermal throttling. Essential to prevent latency spikes during YOLOv8/v11 object detection and SLAM.
	DC Input Voltage	20 V	Ideal voltage supplied by the proprietary adapter, simplifying deployment and safety approvals.
	Operating Current (at 114 W)	5.7 A (114 W/20 V)	Remains safely within the upper current draw constraint of 10 A.
	Adapter Input Req.	133.33 W (120 W/0.90)	Accounts for an assumed 90% adapter efficiency loss that the upstream source must supply.
Depth Camera	Power Draw (Est.)	3.5 W (5 V $\times$ 0.7 A)	Calculated at maximum current constraint (100% efficiency assumed).

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Table 12. System Power Budget and Engineering Rationale (Continued)

Item(s)	Parameter / Metric	Value / Calculation	Engineering Design Choice Rationale
LiDAR	Power Draw (Allocated)	2.5 W	Allocated remaining power from NUC output (0.5 A at 5 V). Meets minimum operation current.
LiDAR + Camera	Peripherals Source	Direct NUC USB-A	Both run from NUC's 5V USB 3.2 Gen 2 ports (1.5 A capacity), ensuring adequate current.
NUC + Peripherals	Total Energy Demand	69.96 Wh (120 W $\times$ 0.583 h)	Calculated on a continuous basis to support uninterrupted obstacle detection, SLAM, and visual processing.
	Total Energy (Input)	77.73 Wh (133.33 W $\times$ 0.583 h)	Includes adapter loss (90% efficiency). Determines true upstream power required for reliable delivery.
Manipulator	Power Draw (Assumed)	60 W	Assumed at 100% efficiency (12 V at 5 A), consistent with DC servo motors in this class.
	Adapter Input Req.	66.67 W (60 W/0.90)	Accounts for 90% adapter efficiency. Proprietary adapter used for safety.
Gripper	Power Source	5 V from Manipulator	No separate source needed. Actuation occurs only when servo motors are idle to ensure wattage availability.
Manipulator + Gripper	Total Energy Demand	34.98 Wh (60 W $\times$ 0.583 h)	Continuous basis calculation to provide a conservative budget covering potential excess duty cycles.
	Total Energy (Input)	38.87 Wh (66.67 W $\times$ 0.583 h)	Includes adapter loss (90% efficiency). Determines true upstream power required for reliable delivery.

4.4 Power Budget Analysis

The detailed power budget breakdown is presented in Table 13.

Table 13. Power Budget Breakdown

Source Power (W)	Adapt. Eff. (%)	Adapt. Out (W)	Component	Comp. Eff. (%)	Comp. Draw (W)	Volt. (V)	Curr. (A)	Time (h)	Energy Comp. (Wh)	Energy Source (Wh)
133.33	90	120	ASUS NUC	95	114	20	5.7	0.583	66.46	
			Intel Depth Camera	100	3.5	5	0.7	0.583	2.04	77.73
			SLAMTEC LiDAR	100	2.5	5	0.5	0.583	1.46	
66.67	90	60	myCobot 280pi (Arm+Grip)	100	60	12	5	0.583	34.98	38.87

Note:

1. Energy Required for Component Operation = Expected Component Power Draw $\times$ Duration of Mission
2. Energy Required from Source = Power Supplied from Source to Adapter $\times$ Duration of Mission

The Leo Rover base station battery has a capacity of 5800 mAh, a nominal voltage of 11.1 V DC, and a maximum output current of 8 A. Based on these specifications, the estimated minimum continuous operating time is 1 hour.

4.5 Power Source Constraints

Supply Constraints

- **NUC supply requirement:** Must deliver 133.33 W AC to the ASUS NUC adapter, which converts to $\approx$ 20 V DC at $\approx$ 6 A. This supports the NUC, Intel RealSense depth camera, and SLAMTEC LiDAR.
- **Manipulator supply requirement:** Must deliver 66.67 W AC to the Elephant Robotics myCobot 280 Pi adapter, which converts to 12 V DC at 5 A for servo motor and gripper operation.
- **Combined AC input current requirement (230V standard, 50Hz):**
 - NUC and peripherals (120 W from 133.33 W) require ≥ 0.58 A.
 - myCobot manipulator and gripper (60 W from 66.67 W) require ≥ 0.29 A.
 - Total minimum supply current: $0.58 + 0.29 = 0.87$ A at 230V AC, 50Hz.
- **Energy capacity:** Must be rated at $\geq 77.73 + 38.87 = 116.6$ Wh to sustain operation for the full mission duration.
- **Safety and compliance:** Must provide stable AC output to meet adapter input requirements, ensuring safe conversion, simplified deployment, and expedited safety laboratory approvals.

Weight Constraints

The rover's payload capacity is constrained by an upper limit of 8.00 kg, which represents an acceptable performance threshold and serves as a compromise between the nominal capacity of 5 kg and the degraded capacity of 10 kg. Within this limit, the combined mass of all known components including the onboard computing unit (NUC), depth camera, LiDAR sensor, manipulator, gripper, adapters, and wiring totals 2.63 kg. An additional approximately 1.20 kg is expected for the payload sled, which includes the block and block storage unit. Subtracting these values from the overall payload limit establishes a maximum allowable battery mass of 4.17 kg. This constraint ensures that the rover maintains the desired operational performance level while adhering to the defined payload capacity.

Dimensionality Constraints

The allocated area for fixing the battery within the rover base is restricted to 18.34 cm in width and 15.3 cm in length. The battery dimensions must conform to these limits, with any permissible overhang strictly controlled to ensure that the rover's center of mass remains balanced. In particular, the battery must not introduce a rearward bias that would render the rover back-heavy.

4.6 Pugh Matrix for Power Source Selection

- **Energy Requirements Compliance** – Implementation meets all specified energy sustainability thresholds (At least 200 W and 116.6 Wh).
- **Weight Requirements Compliance** – Implementation adheres to maximum allowable weight constraints (< 4.17 kg).
- **Dimensionality Requirements Compliance** – Implementation conforms to the prescribed spatial constraints and defined margins (approximately $18.34\text{ cm} \times 15.3\text{ cm}$).
- **Stakeholder Requirements Compliance** – The solution meets stakeholder requirements (preference for non-tethered solutions).
- **Quality of Implementation** – The degree to which the solution achieves intended functionality reliably.
- **Cost of Implementation** – Total expenditure required for design, prototyping, and deployment.
- **Implementation Complexity** – The relative difficulty of integrating the solution into existing systems.
- **Time for Research and Implementation** – The duration required to complete feasibility studies, prototyping, and deployment.
- **Time for Approval** – The expected period for institutional validation.
- **Safety Compliance** – Conformance with all applicable safety standards and risk-mitigation protocols.

Following evaluation of the available options in Table ??, the Anker Solix C300X satisfies all defined criteria while presenting the fewest compromises. Accordingly, it has been selected as the default power source.

Table 14. Pugh Matrix for Power Source Selection

Criteria	Wt.		Current	Alt. 1		Alt. 2		Alt. 3		Alt. 4		
			Anker Solix	Panana Station		Jackery Explorer		Direct Tether		Custom Source		
	B	W	B	W	B	W	B	W	B	W		
Energy Requirements Compliance	5	5	25	3	15	4	20	5	25	3	15	
Weight Requirements Compliance	5	3	15	5	25	5	25	5	25	5	25	
Dimensionality Req. Compliance	4	4	16	5	20	3	12	5	20	4	16	
Stakeholder Req. Compliance	5	5	25	5	25	5	25	1	5	5	25	
Quality of Implementation	4	5	20	4	16	5	20	5	20	3	12	
Cost of Implementation	2	4	8	5	10	4	8	5	10	5	10	
Implementation Complexity	3	5	15	5	15	4	12	3	9	2	6	
Time for Research & Impl.	3	5	15	5	15	5	15	5	15	2	6	
Time for Approval	4	4	16	4	16	4	16	5	20	2	8	
Safety Compliance	5	5	25	3	15	5	25	1	5	2	10	
Total Weighted Score			180		172		178		154		133	

Note:

- **Abbreviations:** **Wt.:** Weight (1–5); **B:** Base Score (1–5); **W:** Weighted Score; **Alt.:** Alternate Solution.
- **Calculation Logic:** Weighted Score (W) = Criteria Weight $\times$ Base Score (B). Total Score = Sum of all W .

4.7 Power Connection Diagram

The specific power connection is shown in Figure 7.

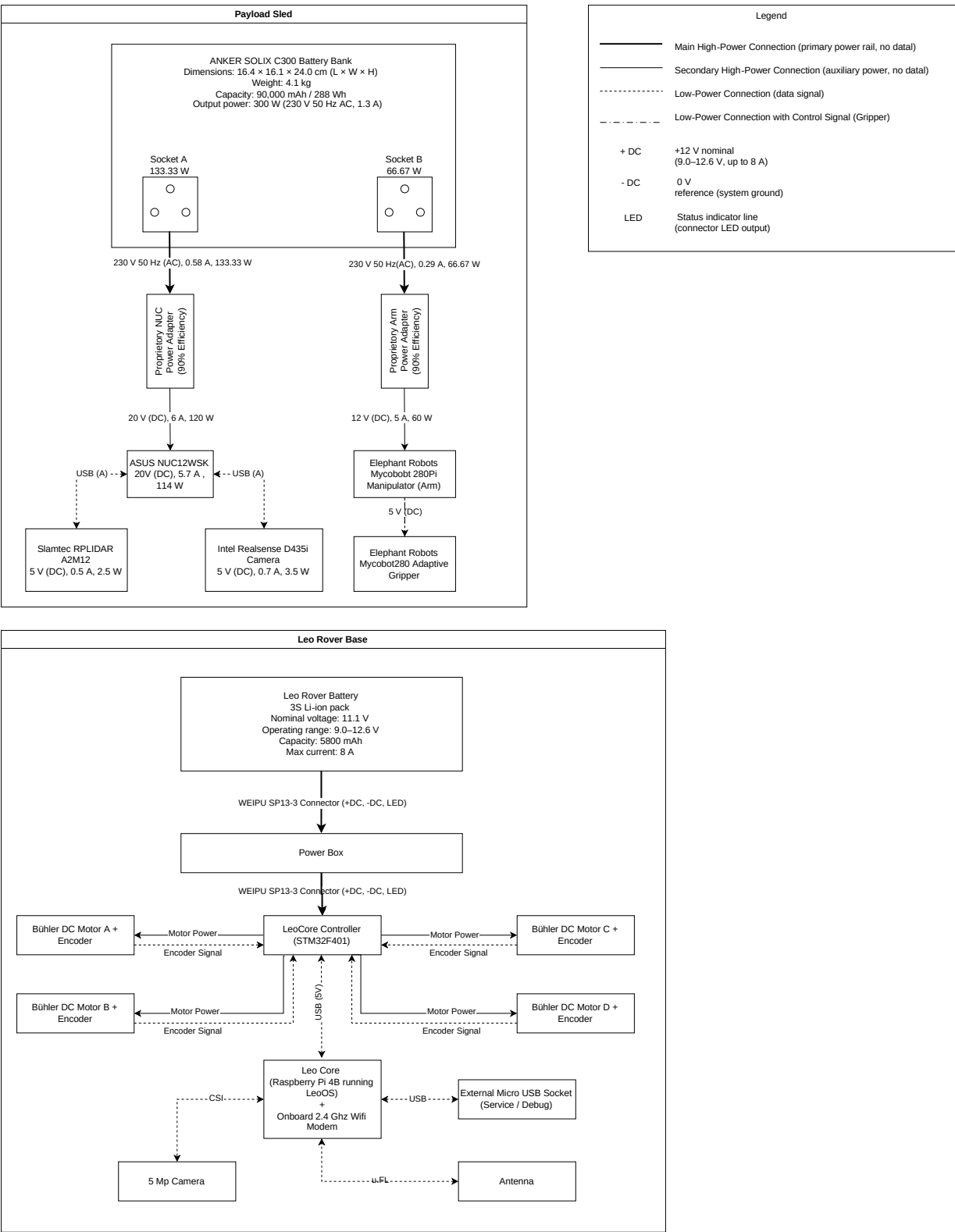


Fig. 7. Power Connection Diagram

5 Software Design

5.1 System Software Design Block Diagram

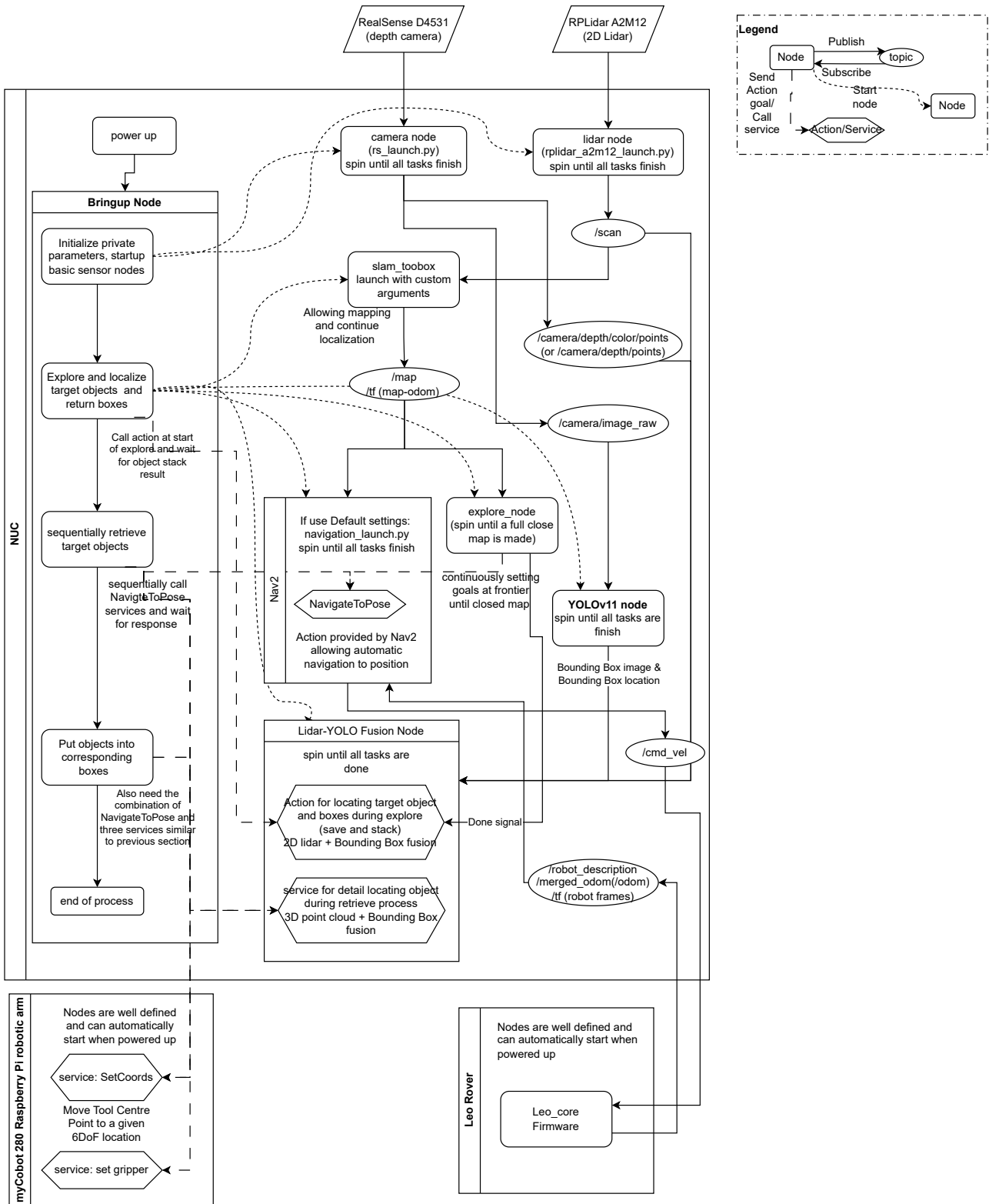


Fig. 8. System Software Design Block Diagram.

Note: Well known predefined nodes only show the used publish topics and subscribe topics; detailed algorithm logics not be shown.

As shown in Figure 8, the system workflow is organized into three main phases: mapping and localization, object retrieval, and final sorting. By building a central bring up node to control the general workflow, and utilizing predefined and custom nodes with actions and services, enabled efficient process management and clear state transitions. By structuring Nav2 navigation actions, exploration triggers, and fusion-based refinement services into a controlled sequence, the system supports flexible interruption handling and smooth switching between exploration, , retrieval , and sorting modes. ”

5.2 Model Evaluation Matrix

To determine the optimal object detection architecture for our robotic platform, we conducted a comprehensive benchmarking analysis on the Intel NUC deployment environment. We evaluated three candidate models—YOLOv8n (baseline), YOLO11n (latest iteration), and RT-DETR-L (transformer-based)—across four critical dimensions: inference speed (FPS), detection accuracy (Confidence), resource efficiency, and feature set. The following evaluation matrix (Table 15) summarizes the weighted performance of each model, highlighting the trade-offs between computational cost and detection precision.

Table 15. Model Evaluation Matrix with Weighted Scores

Metric	Weight	YOLOv8n	YOLO11n	RT-DETR-L
Speed (FPS)	35%	9.0 (58 FPS)	9.0 (58 FPS)	3.0 (13 FPS)
Accuracy (Conf)	30%	7.5 (0.89)	9.0 (0.94)	7.0 (0.87)
Resource Usage	25%	9.0 (Very Low)	9.0 (Very Low)	2.0 (Very High)
Features	10%	8.0	10.0 (Enhanced)	8.0
Total Weighted Score	100%	8.45	9.10	4.45

Quantitative benchmarking on the Intel NUC confirms that YOLO11n is the optimal architecture for our robotic platform. It achieves the highest weighted score by delivering superior real-time performance (58 FPS) compared to RT-DETR-L, while maintaining robust detection accuracy.

5.3 Github Repository

Link to Team Github Repository: https://github.com/longchuanYoRHa/Group12_team_space

6 Analysis

This section analyzes whether the design meets the course requirements using the Requirements Verification Matrix (RVM). Table 16 lists each shall-statement, its corresponding verifiable success criteria, and the analysis and verification plan based on the current mechanical, electrical, and software design.

Table 16. System Requirements and Design Analysis

Req #	Shall Statement	Verification of Success Criteria	Analysis of design
Operation			
1	The robot shall operate within the Test Environment.	Robot operation is not impeded by features of the testing environment.	The design of the robot utilizes the Leo Rover 1.8 platform which has been proven to operate effectively in the lab environment. The mechanical design has not considered dynamic forces due to restricted movement speed and level area. No consideration for extreme conditions (heat, water ingress, etc) necessary.
2	The robot shall be capable of restarting the operation.	The robot can be shown to restart the operation after interruption/turn off state.	The restoration workflow supports full restart after interruption. When the system is rebooted or halted, SLAM resets the map, the bringup node relaunches all navigation components, and the autonomous workflow begins again from the initial state. The robot can therefore resume operation reliably without manual reconfiguration.
2.1	The robot shall achieve full operation in <5 minutes.	During testing the robot can be restarted and fully operational within the 5 minute window.	A timer is initiated at program start to monitor the 5-minute operational window. If available time becomes insufficient, the system safely increases navigation speed within the configured velocity limits allowing SLAM initialization, exploration startup, and autonomous movement to complete within the required timeframe.
3	All hardware shall be mounted on the robot platform.	There is no un-mounted hardware on the robot.	The payload deck design allows for all components to be mounted to the robot securely, and honing ingress points should remain attached.

Continued on next page

Table 16 – Continued from previous page

Req #	Shall Statement	Verification of Success Criteria	Analysis of design
Navigation and planning			
4	The robot shall navigate autonomously.	The robot successfully navigates the environment without external influence.	Based on the hardware restrictions, and the support documents from Leo Rover. Currently the system integrates SLAM Toolbox with Nav2, forming a navigation stack that supports real-time mapping, localization, and autonomous motion planning. In addition, by incorporating the exploration node provided in the m-explore-ros2 package the robot is capable of fully autonomous map exploration. The LiDAR positioning on the payload sled allows 360 degrees of unobstructed sensing which has proven to be sufficient in preliminary testing.
4.1	The robot shall avoid obstacles and unintended collisions.	During operation the robot does not impact the test environment walls, obstacles, coloured bins or boxes unintentionally.	Nav2 avoids obstacles by updating costmaps from LiDAR data, inflating obstacles based on the robot footprint, and selecting only collision-free trajectories through real-time trajectory simulation and controller evaluation.
4.2	The robot shall be capable of receiving commands from external controls.	The robot responds accordingly to commands issued from external controls.	The robot accepts external interrupts through ROS2 action and service interfaces. These allow higher-level controllers to issue navigation or manipulation commands, enabling safe task interaction and immediate behavioral reconfiguration during operation.
4.4	Maximum linear velocity shall be limited to ≤ 250 mm/s.	The robot does not exceed 250 mm/s.	Speed limits are enforced by configuring maximum linear and angular velocities in the Nav2 controller parameters, ensuring all generated trajectories and issued cmd_vel commands remain within required safety constraints.
Continued on next page			

Table 16 – Continued from previous page

Req #	Shall Statement	Verification of Success Criteria	Analysis of design
4.5	Maximum angular velocity shall be $<60^{\circ}/s$.	The robot does not exceed $60^{\circ}/s$.	Angular velocity is constrained by setting the Nav2 controller's max_vel_theta parameter, ensuring all planned motions respect the required rotational speed limit and preventing unsafe or excessive turning during navigation.
5	The robot shall locally generate paths for navigation.	The process of path planning is shown to be running locally.	Nav2 computes all global and local paths onboard using SLAM-generated maps and live sensor data, ensuring fully local planning without dependence on external computation or preloaded routes.
5.1	Path planning shall use data from sensor data not pre-programmed global map.	Sensor data is shown to be driving decision making of the path planning.	SLAM Toolbox builds the map in real time from LiDAR inputs, and Nav2 plans paths using these evolving costmaps, ensuring navigation relies solely on live sensor observations rather than predefined maps.
Vision and perception			
6	The robot shall locate the Coloured Objects.	The robot correctly locates the coloured objects within 3D space.	The system fuses 2D bounding boxes from YOLOv11 with aligned depth frames from the Realsense D435i. This allows for the extraction of spatial ROI (Region of Interest) corresponding to specific objects.
6.1	The robot shall determine the 3D coordinates (x,y,z) of the target relative to the camera.	The robot accurately (within a grasp capable margin) provides the relative XYZ coordinates relative to the camera.	Using the camera's intrinsic matrix, the system deprojects pixel coordinates (u, v) and filtered depth values d into 3D camera coordinates (x, y, z). ROI statistical filtering removes outliers to ensure coordinate precision within +/- 5mm.
Continued on next page			

Table 16 – Continued from previous page

Req #	Shall Statement	Verification of Success Criteria	Analysis of design
6.2	The robot shall locate coloured blocks of size > 20 x 20 x 20mm.	Coloured blocks of size 20x20x20mm shall be used in the coloured object testing to measure success.	The Realsense D435i provides sufficient point cloud density at the manipulation range. The YOLO input resolution (e.g., 640x640) ensures that 20mm objects occupy a distinguishable pixel area for reliable detection.
7	The robot shall classify the Coloured Objects.	The robot correctly classifies the coloured objects.	Classification is performed by the YOLOv11 detection head. The model is optimized via OpenVINO and deployed on the NUC's Iris Xe CPU to achieve real-time inference (>10 Hz) with high confidence scores.
7.1	The robot shall classify coloured blocks of size > 20 x 20 x 20mm.	Coloured blocks of size 20x20x20mm shall be used in the coloured object testing to measure success.	The neural network is trained on a dataset containing small objects. Feature pyramid networks (FPN) within the YOLO architecture enhance the model's ability to extract semantic features from small (20mm) targets.
7.2	The robot shall tolerate lighting variation including shadows and darkening shades of the same nominal colour.	The robot can be shown to correctly identify the bins colour under the natural lighting conditions in the lab.	The training pipeline includes aggressive HSV augmentation (Hue, Saturation, Value jitter) and random brightness/contrast adjustments. This forces the model to learn color invariance, ensuring robustness against shadows and lab lighting shifts.
7.3	The robot shall locate & classify at a distance >75mm.	The robot provides accurate positional data and correctly identifies the objects colour whilst positioned adjacent to the coloured objects mounting box.	The Realsense D435i is configured to operate effectively from its minimum sensing distance (~0.28m) outwards. For distances >75mm (specifically in the arm's workspace), the depth accuracy is optimal for grasping planning.

Continued on next page

Table 16 – Continued from previous page

Req #	Shall Statement	Verification of Success Criteria	Analysis of design
8	The robot shall locate & classify the Coloured Bins.	The robot provides accurate positional data and correctly identifies the bins colour during testing.	Bins are detected as distinct classes (e.g., class: red_bin). Their 3D centroids are transformed via the ROS TF Tree into the global map frame, allowing the navigation stack to plan approach paths for depositing objects.
Object manipulation			
9	The robot shall manipulate the Coloured Objects.	The robot successfully manipulates the Coloured objects in the desired way.	The manipulator subsystem transforms its pose into the manipulator base coordinate system via a transformation tree to execute a deterministic grasping process. The manipulator controller strictly enforces joint limits and position accuracy, while the motion planner generates collision-free joint trajectories.
9.1	The minimum size of the Coloured Objects to be manipulated shall be 20 x 20 x 20mm.	The robot successfully manipulates objects of the size specified.	The manipulator's repeatability accuracy (± 0.5 mm), gripper opening range (20-49 mm), gripping force, and control precision (positioning accuracy ± 1 mm) meet the requirements for grasping and transporting objects of this minimum size.
9.2	The robot shall manipulate at a height of 200mm above the floor.	The robot is shown to manipulate objects on top of the coloured block plinth and at the same height to the coloured deposit bins.	The manipulator's kinematics and joint limits enable the end effector to perform approach, grasp, and place motions at 200 mm. Visual depth perception provides pose correction for final alignment.
9.3	The maximum weight of the target blocks shall be 50g.	When calculated, none of the coloured blocks shall weigh more than 50g.	The payload capacities of the myCobot 280 Pi and gripper are 250g and 150g respectively. The gripper itself weighs 110g. Considering the weight of the building blocks, the upper limit selected is 50g.

Continued on next page

Table 16 – Continued from previous page

Req #	Shall Statement	Verification of Success Criteria	Analysis of design
10	The robot shall transport the Coloured Objects.	The robot will successfully carry coloured objects between given locations.	The navigation stack (SLAM Toolbox + Nav2) generates collision-safe paths under local map and obstacle expansion constraints. The vehicle controller ensures linear and angular velocity limits (< 250 mm/s, < 60°/s) to maintain transport stability. After grasping, the robotic arm places objects onto the payload pallet and retrieves them again upon returning to or approaching the drop point for precise placement.
10.1	The robot shall transport a minimum of 1 Coloured Object.	The robot will show it can transport a single coloured object inside the test environment.	Preliminary testing has shown that the gripper can successfully grasp and hold a coloured object, and retain it whilst in motion. This is how the design shall transport a single coloured block. If required a 3 compartment coloured object container can be mounted to the existing payload deck to enable transport of multiple objects.
11	The robot shall deposit the Coloured Objects into the corresponding Coloured Bins.	For 3 successive trials, the robot will deposit the Coloured object into the correct bin.	The system will automatically relocate and retry within the permitted number of retries. All events (success/failure, retry count, timestamp) are logged to demonstrate the repeatability and reliability of three consecutive successful attempts.
Safety			
12	The robot shall be capable of being manually handled.	The robot shall be shown to be safely picked up and placed into the test environment.	The mechanical design and positioning of components do not preclude the ability to safely handle the robot. Consideration has been given to not impede existing lifting points and to lifting strategy.

Continued on next page

Table 16 – Continued from previous page

Req #	Shall Statement	Verification of Success Criteria	Analysis of design
12.1	The combined weight of the robot and the payload shall be <20kg.	When calculated the sum of weights of all the components and the payload shall be less than 20kg.	The combined weight of all known components is 2.38 kg, the selected battery weighs 4.10 kg, and the payload sled with block storage is expected to add approximately 1.20 kg. The rover base contributes 6.50 kg, bringing the total estimated weight to about 14.48 kg. This remains below the target weight constraint, ensuring compliance with performance requirements.
13	The robot shall be able to be stopped in the case of unexpected behaviour.	Emergency shutdown button demonstrated.	The Leo Rover and C300 have physical power buttons that cut power when pushed, and will serve as emergency shutoff. This is deemed acceptable risk given the controlled environment.
13.1	The robot shall be capable of being issued a stop command wirelessly.	The robot will show the ability to be stopped remotely.	The robot can be stopped wirelessly via Bluetooth or an existing SSH connection. A remote operator uses a software interrupt that cancels all active ROS2 actions and commands the bringup node to publish a zero-velocity vector, ensuring rapid and reliable termination of ongoing navigation or manipulation tasks.
13.2	The robot shall have a form of easily accessible emergency shutdown button or switch.	Emergency stop button is present, indicated, and accessible.	The addition of red tape (or some other visual indicator) to the Leo Rover and C300 power buttons shall provide visual indication of power cutoff.

Power

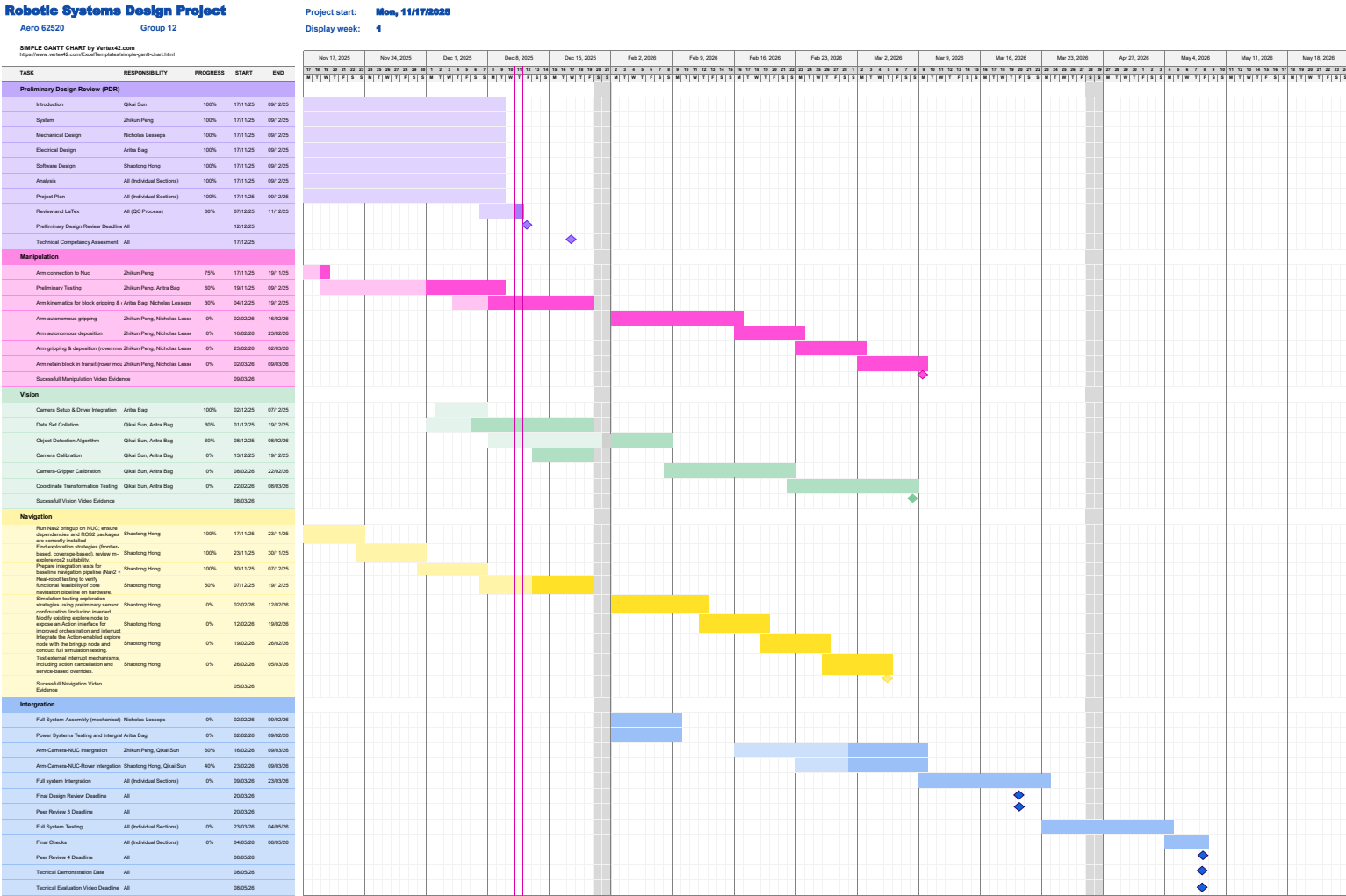
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Table 16 – Continued from previous page

Req #	Shall Statement	Verification of Success Criteria	Analysis of design
14	Electrical design and maintenance shall comply with lab risk assessment.	The power supply and batteries can be shown to comply with the risk assessment by design and when inspected.	The chosen power supply delivers the required wattages to the NUC, peripherals, manipulator and gripper with ample margin for the demonstration, while the Leo Rover base remains powered by its OEM battery. All power components and OEM adapters/cables are selected from reputable suppliers and manufacturers who perform their own testing and quality inspection, minimizing the risk of short circuits, voltage transients, or electrical noise, and causing thresholds follow the manufacturer's specifications. During the 20-minute mission this rover is expected to complete the specified tasks without depleting the battery reserve without short circuits or electrical damage to components, and maintain full performance with no degradation caused by insufficient power.

7 Project Plan

The detailed project schedule, milestones, and task allocations are illustrated in the Gantt chart in Figure 9.



8 Conclusion

This report outlines the primary design decisions and engineering rationale during the Preliminary Design Review (PDR) phase. System requirements were decomposed based on course requirements and the requirements matrix, with specific implementation plans and design specifications provided for the mechanical, sensing, navigation, and power subsystems.

Appendices

A Appendix A: CW1 Workplace Charter

CW1 Workplace Charter

Statement of Purpose

This workplace charter sets out the framework for how Team 12 will collaborate, make decisions, and manage responsibilities. It defines standards for workflow, time and workload management, reporting, scheduling, inclusion, conduct, and safety. It guides interactions within the team and with university staff and provides a process for conflict resolution. It is an obligation that all team members subscribe to and will be used throughout the project to support fair, safe, and effective teamwork. This charter is a living document and may be revised as the project evolves. All changes will be recorded in the version history and communicated to the team.

Principles and Commitments

As a team, we will strive to uphold professionalism, respect, and integrity in all aspects of our work.

ED&I - We will commit to fostering an environment grounded in equality, equity, diversity, and inclusion. We will ensure that every member will have a voice, their opinion will be valued, and they feel fully included. We will champion the protection of all legally recognised characteristics—sex, gender identity, disability, ethnicity, religion or belief, sexual orientation, marital or transgender status, age, nationality, and neurodiversity—and reinforce the safety and wellbeing of every individual.

Collaboration - We will leverage each member's strengths, support one another's growth, and prioritise collective success over individual recognition. We will uphold transparency in workload and time management, meet deadlines reliably, and schedule meetings with fairness and flexibility. We will communicate openly, listen actively, and respond respectfully.

Safety - We will follow all safety protocols without compromise, and ensure our design decisions reflect reliability, performance, and feasibility.

Professionalism - We will maintain professionalism in all interactions—with teammates, instructors, GTAs, other teams, and the wider university—and address conflicts constructively. In cases of serious misconduct, including harassment, discrimination, or gender-based violence, we will escalate directly to the appropriate university department. We will hold ourselves accountable, seek feedback, share knowledge, and continuously improve.

Through these commitments, we will build a safe, inclusive, and high-performing environment where collaboration, innovation, and technical excellence will thrive.

Team Rules

Section	Focus Area	Guidelines
Working Hours	A. Working Hours	i. Working hours are between 10am – 6pm, Mon–Fri, . ii. 5h asynchronous time per week. iii. Other times in urgent situations: safety risks, personal injury, severe equipment damage, irreversible delivery risks, etc.
Lab Attendance and Participation	B. Attendance	i. Mandatory for all scheduled lab sessions. ii. Notify team 24 hours in advance if unable to attend.
	C. Preparation	i. Arrive 15 minutes early to collect equipment and set up. ii. Complete pre-lab work (mechanical design and software control).
	D. Session Conduct	i. Follow all lab health and safety regulations. ii. Comply with all direction given by GTAs and staff.
	E. Rotating Responsibilities	i. Bring rover kit and return to storage. ii. Cleanup workspace. iii. Charge rover batteries.
	F. End-of-Session Protocol	i. Finish work at least 15 minutes before the lab session ends. ii. Use final 10 minutes to tidy up and verify inventory. iii. Use final 5 minutes to return the rover kit to storage. iv. Return the rover battery to the GTAs after each session.
	A. Response Times	i. General: within 24 hrs (work hours). ii. Urgent: as soon as possible, during work hours only.

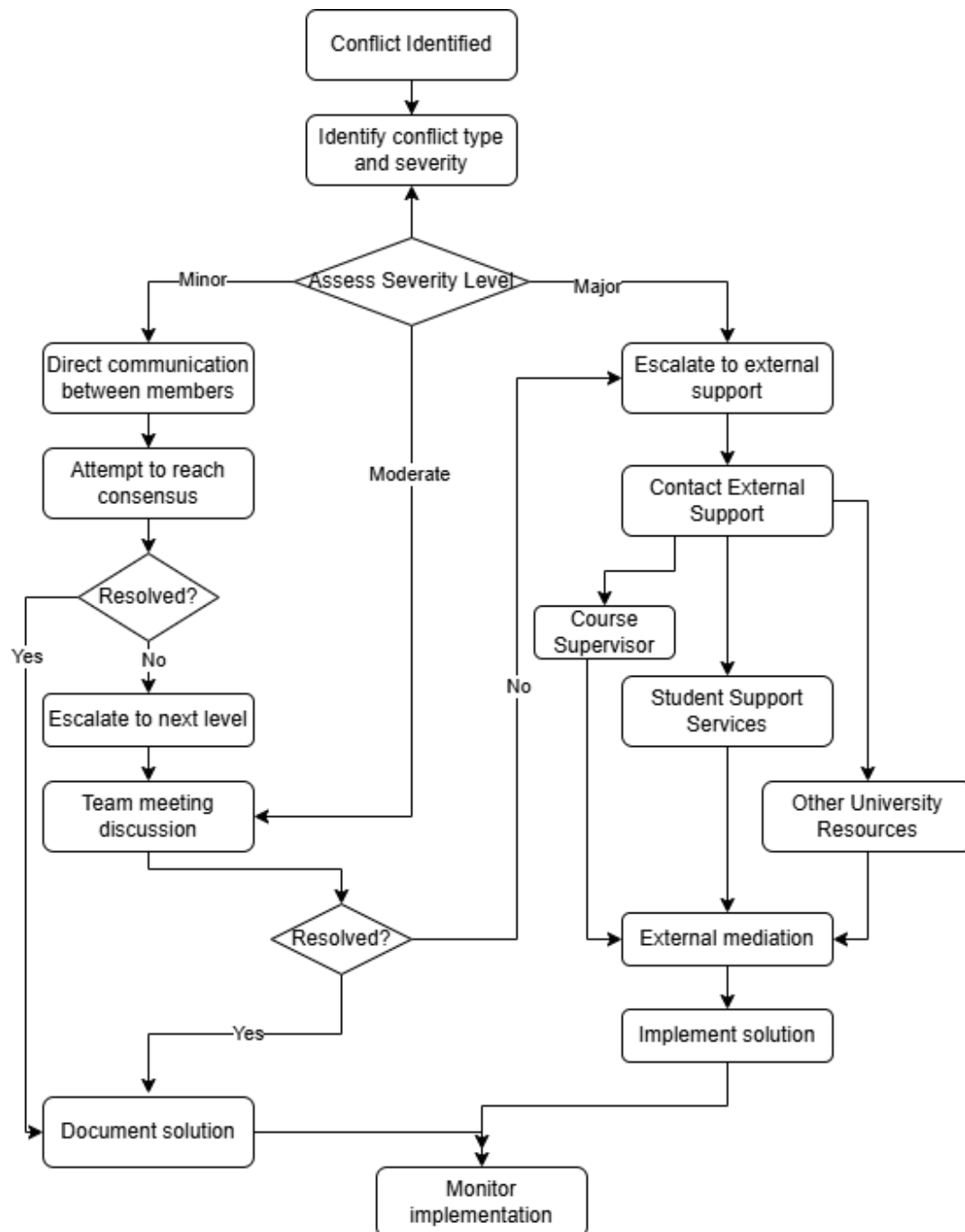
Communication and Language Standards		iii. Weekend: no response expected unless urgent.
	B. Channels	i. University Email and Microsoft Teams for formal communications, project discussion, meeting logs. ii. GitHub for code documentation, and version control.
	C. Language Use	i. Communicate in English across all platforms and meetings. ii. Support non-native speakers and encourage clarification. iii. Maintain clear, respectful, and inclusive language.
Decision-Making Process	A. Minor Decisions	i. Decide tool choices, formatting, and scheduling within sub-teams. ii. Inform full team via communication channels.
	B. Major Decisions	i. Discuss design choices, strategy, and major purchases in team meetings. ii. Aim for consensus with majority vote (minimum 80% team presence). iii. Document all decisions with rationale.
Work Allocation and Resource Management	A. Task Assignment	i. Assign tasks considering skills, workload, and learning opportunities. ii. Record tasks in GitHub-based shared task tracker or Teams channel post.
	B. Deadlines and Progress	i. Set internal deadlines 48 hours before external ones. ii. Share progress updates regularly (at significant milestones). iii. Notify team as soon as possible if unable to meet a deadline.
	C. Shared Resources	i. Borrow only tools and materials outside the provided kits. ii. Return borrowed items on time and report any damage or loss ASAP. iii. Keep a record of robotic kit inventory.
Quality Standards and Documentation	A. Code and Design	i. Follow agreed coding style guide (e.g., PEP 8 for Python). ii. Comment all code and include a README. iii. Submit code only after testing and peer review.
	B. Documentation	i. Log weekly (individual): work, time, issues, next. ii. Document all design decisions with rationale. iii. Organize Python and ROS files with clear naming conventions.
	C. Version Control	i. Write descriptive commit messages. ii. Require at least one approval for pull requests.
Wellbeing and Conflict Management	A. Wellbeing	i. Respect work-life balance; avoid working beyond work hours. ii. Support team members facing personal difficulties. iii. Take regular breaks during long work sessions to prevent burnout.
	B. Conflict Management	i. Adhere to Daily Activity Conflict Avoidance practices (below). ii. Comply with Conflict Resolution Flow Chart (page 3).

Daily Activity Conflict Avoidance

To avoid formal grievances, the Team will follow these daily practices:

- i. Three stand-ups a week to quickly note blockers, clarify ownership, reducing misunderstandings or duplicated work caused by delayed information.
- ii. Teams posts and a risk register visualise task status and potential issues, enabling the team to act before problems escalate.
- iii. Change request templates and a decision log prevent “verbal changes” or undocumented decisions, reducing later disputes.
- iv. Weekly Team Retrospectives with a closed feedback loop record mistakes and implement improvements, reducing recurring conflicts.
- v. Always talk first policy followed by tiered escalation process (private first, then team mediation, then tutor) provides a clear, fair order for raising concerns and prevents emotional public disputes.
- vi. Serious incidents should be reported immediately, and evidence preserved to protect affected individuals and support formal School procedures.

Conflict Resolution Flowchart



Links:

Report and Support - <https://www.reportandsupport.manchester.ac.uk/> - Hub for bullying, harassment, discrimination, gender based violence, hate or micro-aggressions, or safeguarding.

Student Support - <https://www.studentsupport.manchester.ac.uk/taking-care/> - Information on things you can do to look after your wellbeing and contacts in case of a crisis.

B Appendix B: CW3 Design Requirements Analysis



AERO62520 Coursework-3 Design Requirements Analysis

Master of Science in Robotics
in the Faculty of Science and Engineering

Date of submission

November 2025

Submitted by

Team 12

School of Engineering

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Declaration of originality

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1 Problem Statement

Growing populations and rising demand for faster, safer production and services have outpaced the available skilled workforce, driving industries to automate repetitive tasks and streamline operations. Modern Autonomous Mobile Robots or AMRs have demonstrated the ability to serve as compact, affordable platforms that embody the essential capabilities of modern autonomy—mapping, self-navigation, dynamic obstacle avoidance, object recognition and manipulation—while providing practical testbeds for systems integration, evaluation and iterative development.

AMRs of this class already show clear value across multiple sectors. In warehousing and logistics, they automate intra facility transport, tote transfer and picker delivery to cut walking time, smooth congestion with dynamic routing and improve throughput and inventory accuracy. For last mile and campus delivery they provide frequent, low-cost micro deliveries on short routes while operating safely in mixed pedestrian areas. In agriculture they perform row by row scouting, environmental sensing and tray retrieval to increase sampling cadence and crop health data quality. In facilities and maintenance, they conduct autonomous patrols that map spaces, detect thermal or visual anomalies, log issues and ferry small tools or parts. Additional applications include healthcare and laboratories for internal transport of samples and supplies, retail and hospitality for restocking and in store delivery, construction site logistics for short range transport in changing environments, and security monitoring through routine autonomous rounds.

Recognizing the growing importance of AMRs, the University of Manchester has offered a project to develop an autonomous rover for mapping, navigation, obstacle avoidance and object retrieval. The rover must reliably locate, grasp and return a standardized block within a target time window, navigate and avoid static and dynamic obstacles in randomized, unmapped indoor arenas without preloaded maps, and make effective use of available onboard hardware—primarily sensors and actuators—while integrating a power system that sustains all components for the duration of test sessions and meeting laboratory safety via documented risk assessment. The platform must provide onboard mapping and localization, waypoint navigation, real-time obstacle detection and avoidance, object recognition and pose estimation, and repeatable block retrieval with return-to-base or delivery to a designated destination, all delivered under constraints of limited hardware availability, randomized test layouts, strict safety requirements and the project's schedule and resource limits; to mitigate principal risks such as localization drift, grasp uncertainty, sensor noise and power or thermal limits the design should incorporate robust perception and sensor fusion, recovery and re-localization behaviours, compliant grasping with visual/force servoing, conservative power budgeting and active monitoring, together with documented safety controls (emergency stop, speed limits, etc). Success will be assessed against the defined functional and performance requirements using

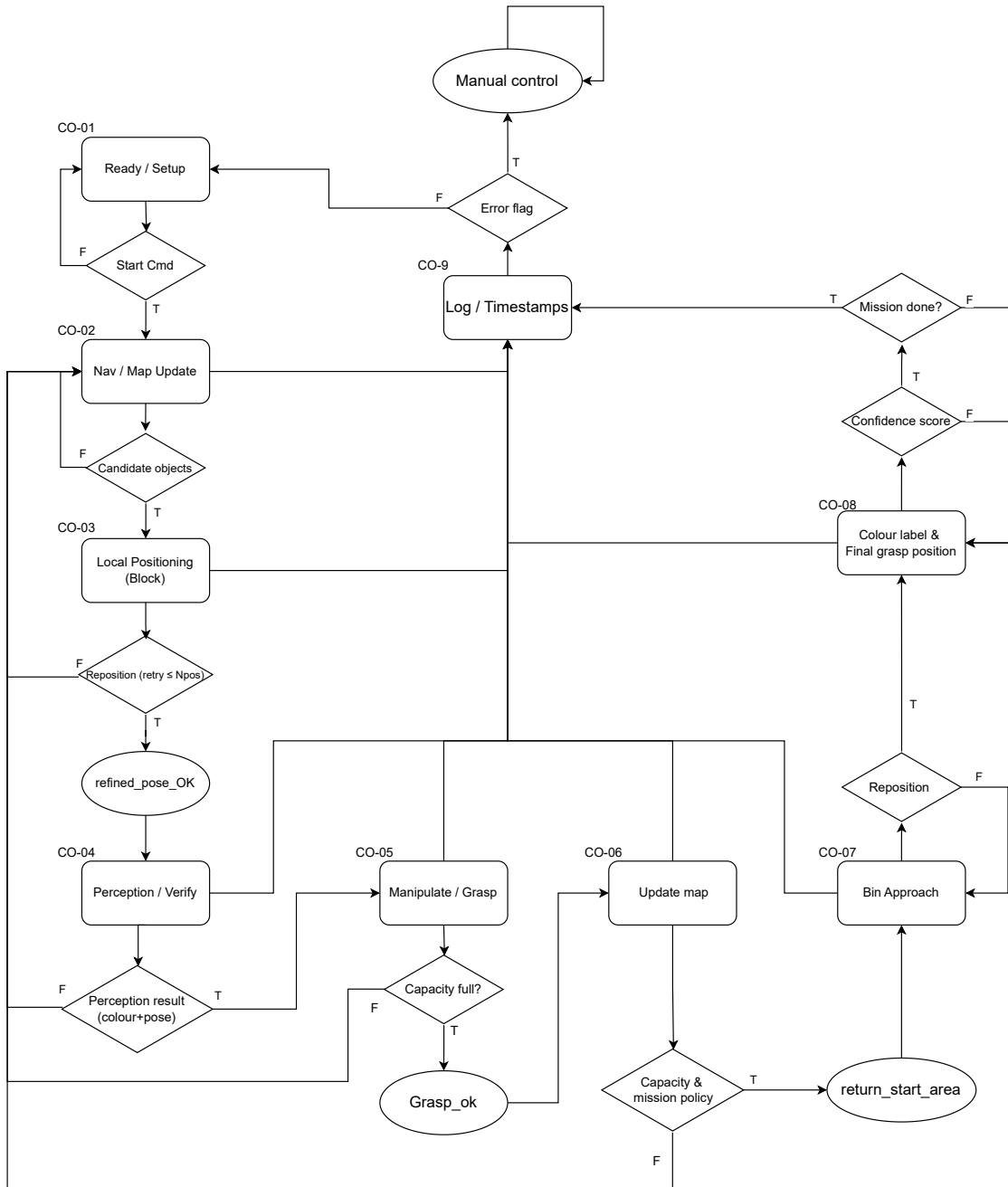
verification metrics with threshold pass/fail values.

In summary, the task is to design, build and validate an autonomous rover that performs repeatable block retrieval missions in unmapped, dynamic environments while remaining safe, robust and modular, producing documented, measurable outcomes that address practical sector needs and advance understanding of systems integration in autonomy.

2 Concept of Operations (CONOPS)

2.1 CONOPS State Transitions

This state-machine diagram Fig. 1 provides a compact mission logic overview, linking all CONOPS phases and the decision gates that move the rover forward or trigger recovery. It answers: (1) Where in the mission loop are we? (2) What condition must be satisfied to advance? (3) What happens on failure or operator intervention?



Note:
T: True
F: False

Fig. 1. State-machine for CONOPS: shows phase flow from CO-01 READY through CO-02 mapping, CO-03 local positioning, CO-04 to CO-06 grasping and map update, CO-07 to CO-09 deposit, plus error handling, timestamps and manual control.

2.2 Start and Setup

Purpose: As shown in Fig. 2, the operator places the rover in the designated Start Area, confirms arena readiness (walls, bins, declared colours, and static obstacle placement), powers sensors and actuators, and configures declared run parameters. Operators confirm sensor health and mission

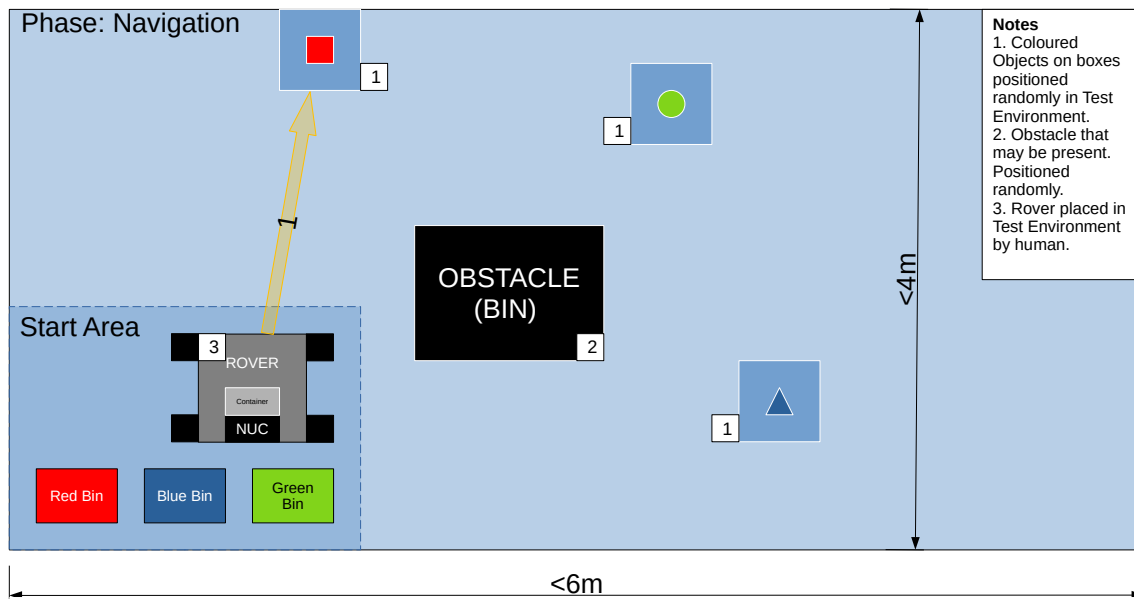


Fig. 2. CO-01 Navigation setup: rover positioned in Start Area beside colour-coded bins; planned route (yellow arrow) toward scattered target zones is shown and a large black rectangle marks a static obstacle.

parameters on the operator interface and issue the start command to begin autonomous operation.

Preconditions: Start Area placement by operator and availability of declared run parameters.

Postconditions: Rover is powered, sensors and actuators pass health checks, run parameters are recorded, and the mission manager reports READY.

Recovery: If any critical sensor or actuator fails health checks, the system alerts the operator and allows sensor restart or manual intervention; if recovery fails within the allowed setup window the run is aborted.

2.3 Autonomous Navigation to Search

Purpose: As shown in Fig. 3, after start, the rover departs the Start Area and performs arena-wide autonomous traversal to search for coloured blocks using onboard LIDAR and camera sensors while respecting arena boundaries and avoiding static obstacles.

Preconditions: LIDAR point clouds, camera images, map/boundary data, and declared colour set.

Postconditions: Candidate object detections with coarse positions and an updated local occupancy map.

Recovery: Upon localization drift or odometry failure, switch to visual re-localization or return to the Start Area for manual recovery.

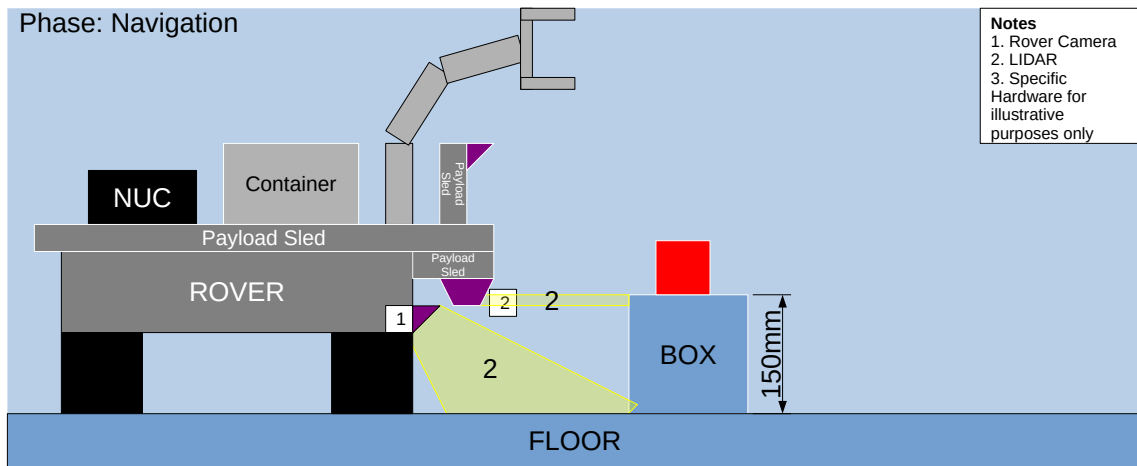


Fig. 3. Navigation-phase diagram for CO-02: the rover uses camera and LIDAR (sensor cones shown) to scan ahead, detect a red coloured object on a box, and update the local occupancy map while avoiding obstacles.

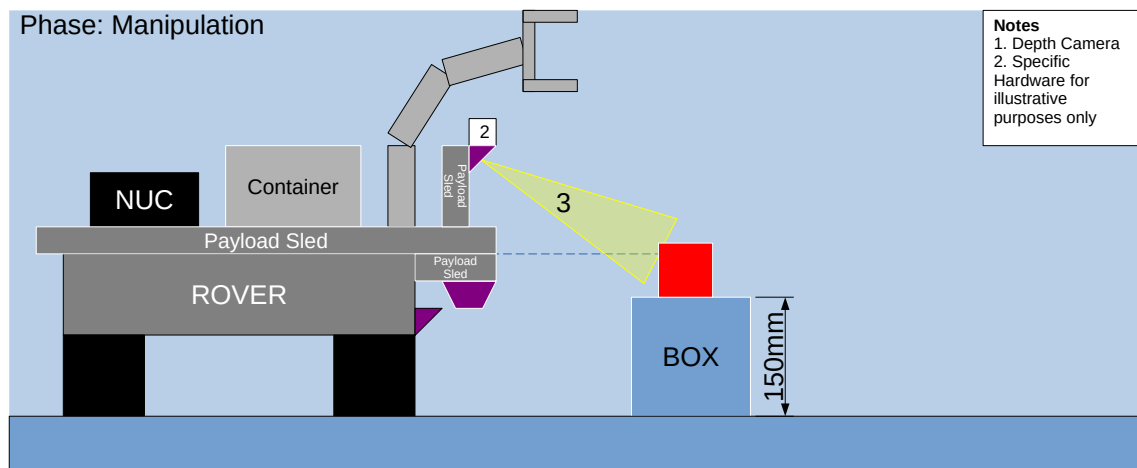


Fig. 4. CO-03 Local positioning: the rover aligns using depth camera and LIDAR (sensor cones) to refine the block pose and attain a collision-safe manipulator approach (300 mm clearance shown)

2.4 Local Positioning at Target

Purpose: As shown in Fig. 4, on detection of a candidate block or bin, the rover executes a local approach routine to position itself adjacent to the target using fine depth/LIDAR alignment to reach a stable, collision-safe pose suitable for perception and manipulation.

Preconditions: Coarse candidate pose, vehicle odometry, and close-range sensor data.

Postconditions: A refined target pose relative to the robot and a verified safe manipulator approach pose.

Recovery: If the rover cannot achieve a collision-free grasping pose within N retries, the target is marked as unavailable, and the rover resumes its search.

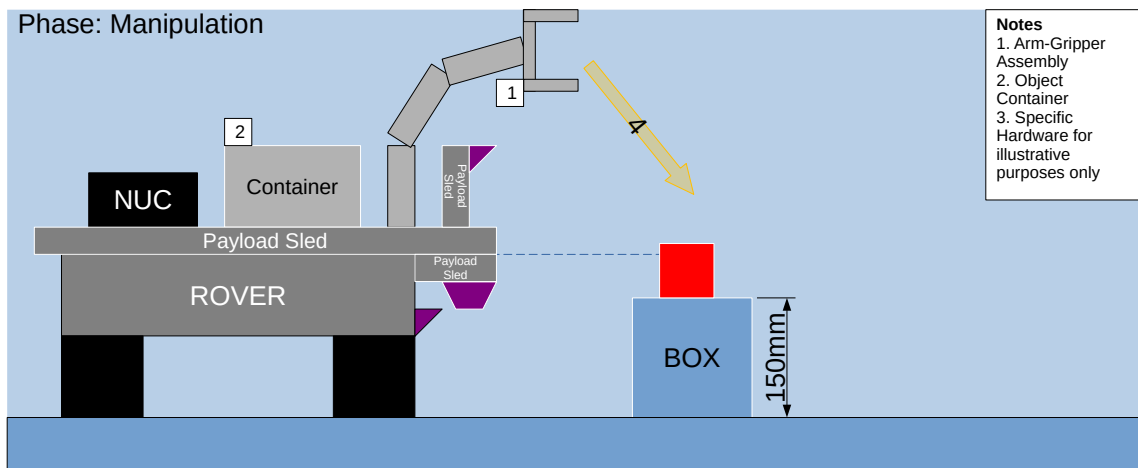


Fig. 5. CO-04 Perception: sensors refine block pose and colour classification; confidence metric confirms readiness for manipulation.

2.5 Perception and Verification

Purpose: As shown in Fig. 5, sensors refine object localization, classify candidate objects against the declared colour set, and publish pose plus confidence for downstream decision-making. Perception must confirm the object meets classification and pose-confidence thresholds before manipulation is attempted.

Preconditions: Sensor colour data, candidate pose and declared colour labels.

Postconditions: Final manipulation pose, colour label, and confidence score.

Recovery: On low confidence, system classifies as false positive and continues search.

2.6 Manipulation

Purpose: As shown in Fig. 6, the manipulator aligns the end effector to the final manipulation pose, executes end-effector sequence, verifies success with onboard sensors (force, tactile, or visual), and secures the object for transport. If verification fails, recovery behaviours are executed.

Preconditions: Final manipulation pose and end-effector parameters from perception.

Postconditions: End-effector success/failure flag and object deposited in payload location.

Recovery: On end-effector failure perform safe retreat, adjust approach or end-effector parameters, and retry up to a configured attempt limit, if still unsuccessful mark target as failed.

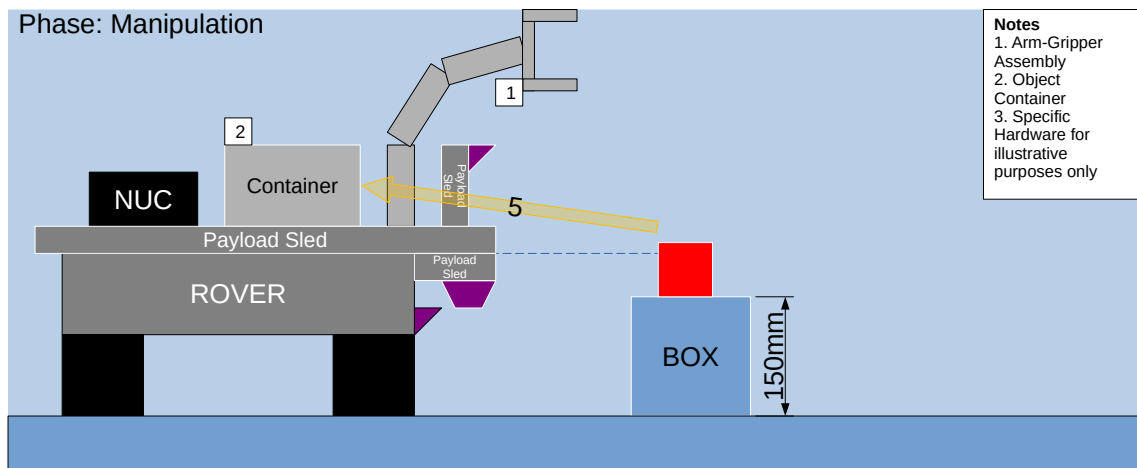


Fig. 6. CO-05 Manipulation: the arm aligns to the final pose and follows the indicated path to grasp the red block on the 150 mm box, then places it into the payload container; success is verified by onboard sensing.

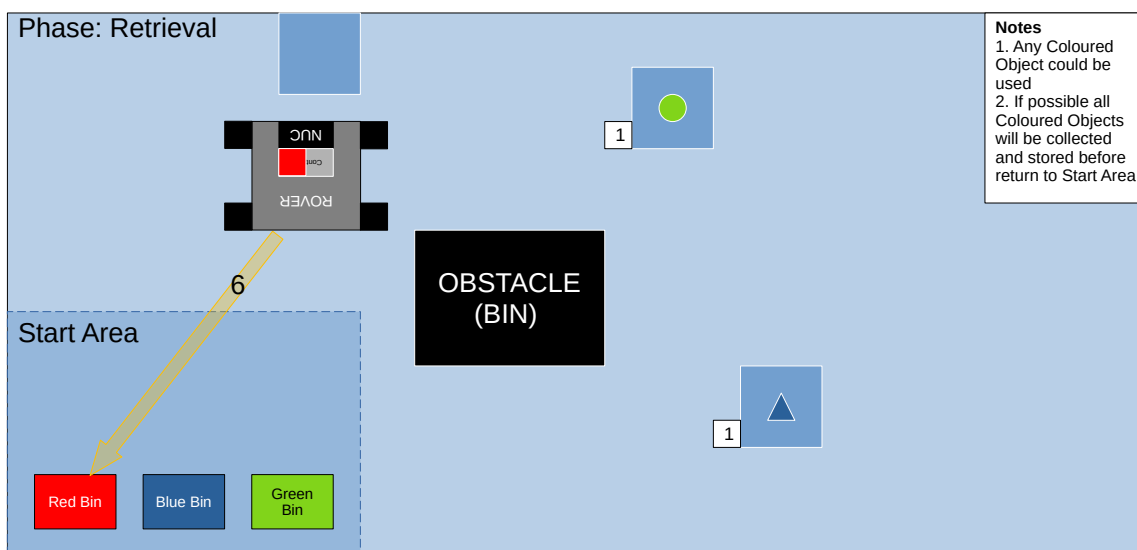


Fig. 7. CO-06 Return to Start Area: the rover follows the indicated path to the Start Area and colour bins while avoiding the central obstacle.

2.7 Robot Returns to Start Area

Purpose: As shown in Fig. 7, based on mission policy (e.g., capacity reached or ordered mode rules), the rover returns to the Start Area or repositions for further retrieval/ deposition tasks.

Preconditions: Storage/capacity state, current map, Start Area pose, mission policy.

Postconditions: Rover pose returns to Start Area or prepared deposition location, mission state update (arrived), and deposition request if applicable.

Recovery: If path blocked, re-plan or choose alternate route; if localization fails, perform visual re localization or request operator assist.

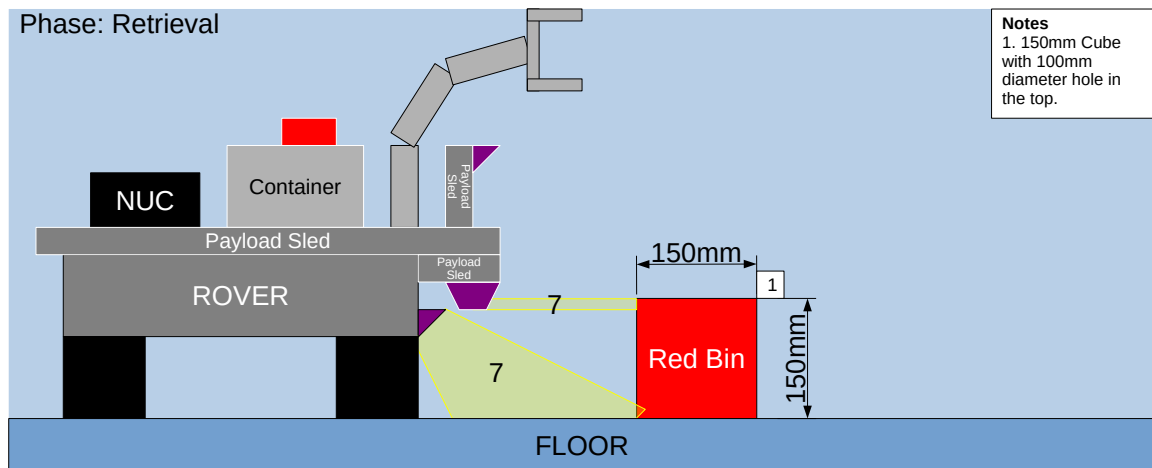


Fig. 8. CO-07 Bin adjacency: the rover halts with a 150 mm standoff to the coloured bin and aligns a clearance-safe approach corridor (shaded) for deposition.

2.8 Robot Positions Adjacent to Coloured Bin

Purpose: The robot uses sensor guidance for navigation and positioning itself beside the target-Coloured Bin to establish a safe approach path for Coloured Object delivery, as shown in Fig. 8.

Preconditions: Bin coarse position, sensor data, odometry.

Postconditions: Approach pose for bin deposition (robot base pose), local map update.

Recovery: If bin not found or occluded, attempt alternate viewpoints or use operator input.

2.9 Deposition into Colour-Matching Bin

Purpose: As shown in Fig. 9, the rover approaches the declared bin on the floor, verifies bin pose and colour, aligns the manipulator with safe clearance from the payload sled/container, and deposits the block; ordered-mode constraints are enforced if active, shown in Fig. 10.

Inputs: Confirmed bin pose and colour, block pose in gripper, deposition plan/payload item info.

Outputs: Deposition success/failure record, updated payload inventory and task counts.

Recovery: On misplacement, drop, or bounce-out, reposition and retry up to the allowed attempts; on persistent failure, log the event and either continue the mission or request operator intervention according to mission rules.

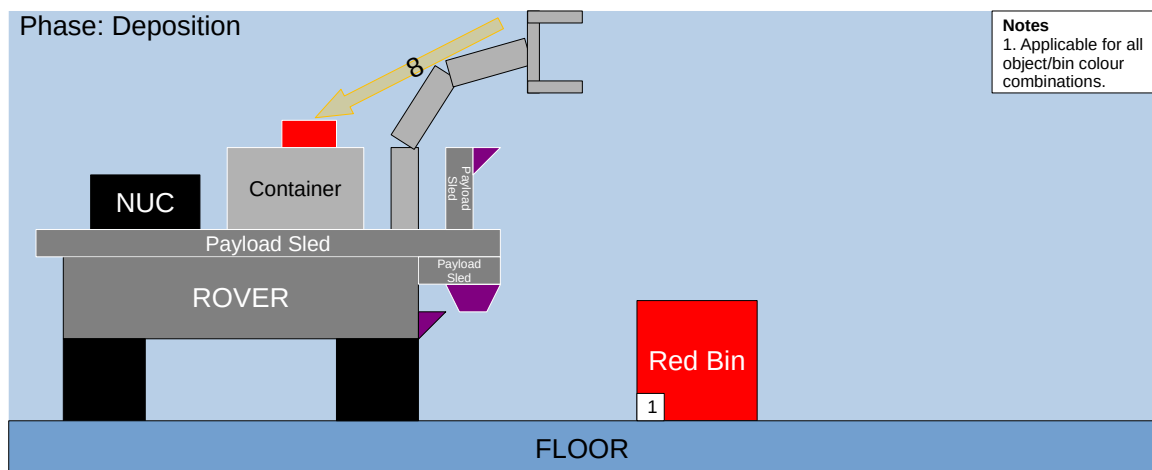


Fig. 9. CO-08 Deposition: rover verifies the red bin, lifts the block from the payload sled/container, and initiates a clearance-safe swing to transfer the block toward the matching floor bin.

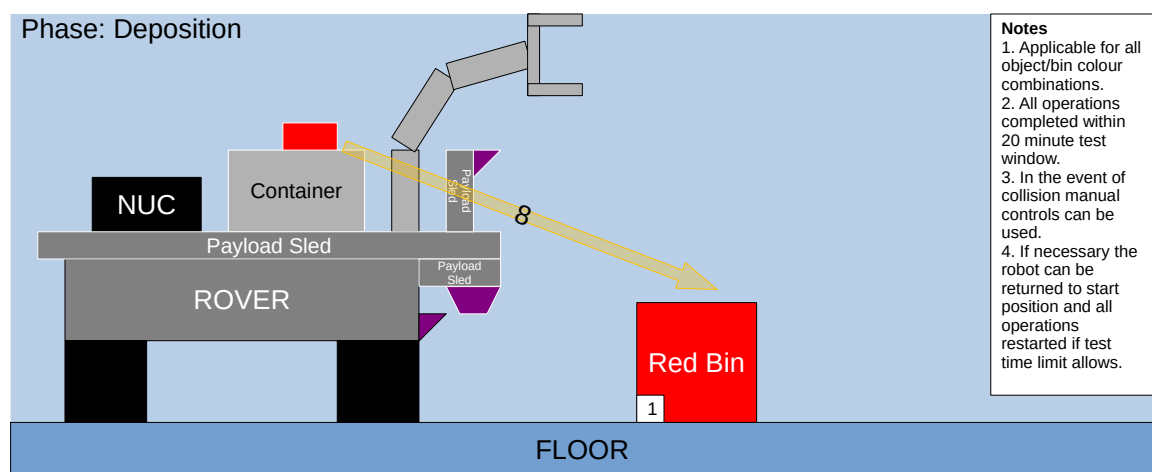


Fig. 10. CO-09 Deposit: the arm follows the indicated path and releases the block into the matching red bin on the floor; success is confirmed and logged, with retry on miss per mission rules.

2.10 Safety, Operator Controls, and Mission End

Purpose: Define safety controls and operator interfaces: users may pause, emergency-stop, resume, or restart via hardware and operator interfaces. In case of collision or unrecoverable fault the system permits manual intervention and restart within the test window. At mission end the system performs safe shutdown and archives logs.

Preconditions: Emergency/pause signals, fault detection, and mission-timer status.

Postconditions: System safe state, archived logs, and restart/terminate decision.

Requirements: Emergency-stop must place drive systems into a safe state and log the event with timestamps.

3 Design Requirements

Table 1. Design Requirements

No.	REQUIREMENT	SOURCE
Operation		
1	The robot shall operate within the Test Environment.	Customer requirements Statement. Stakeholder engagement
1.1	The boundary will be defined by a white wall <0.5m high.	Stakeholder engagement
1.2	The Test Environment will be <4m x <6m.	Collected Data. Stakeholder engagement
1.3	The Test Environment will be level.	Stakeholder engagement
2	The robot shall be capable of restarting operation during testing.	Engineering Judgement
2.2	The robot shall achieve full operation in <5 minutes.	Stakeholder engagement
3	All hardware shall be mounted on the robot platform.	Stakeholder engagement
3.1	Combined weight of payload should be less than 5kg.	Data sheet - Leo Rover 1.8
Navigation and Planning		
4	The robot shall navigate autonomously.	Customer requirements Statement. Stakeholder engagement
4.1	The robot shall avoid obstacles and unintended collisions.	Engineering Judgement
4.2	The robot shall be capable of receiving commands from external controls.	
4.3	External controls will only be used in event of a collision or unexpected behaviour.	Stakeholder engagement
4.4	Maximum linear velocity shall be limited to ≤ 250 mm/s.	Risk Assessment
4.5	Maximum angular velocity shall be $<60^\circ/\text{s}$.	Data sheet - Leo Rover 1.8
5	The robot shall locally generate paths for navigation.	Engineering Judgement
5.1	Path planning shall use data from sensor data not pre-programmed global map.	Stakeholder engagement
Continued on next page		

Table 1 – Continued from previous page

No.	REQUIREMENT	SOURCE
5.2	Path planning should avoid tangling of cables (if present).	Risk Assessment
Vision and Perception		
6	The robot shall locate the Coloured Objects.	Stakeholder engagement, Engineering Judgement
6.1	The robot shall determine the 3d coordinates (x,y,z) of the target relative to the camera.	Engineering Judgement
6.2	The robot shall locate coloured blocks of size $> 20 \times 20 \times 20$ mm.	Collected Data.
7	The robot shall classify the Coloured Objects.	Engineering Judgement
7.1	The colour of the Coloured Objects will be decided by Team 12.	Stakeholder engagement
7.2	The robot shall classify coloured blocks of size $> 20 \times 20 \times 20$ mm.	Collected Data.
7.3	The robot perception shall tolerate lighting variation including shadows and differing shades of the same nominal colour.	Stakeholder engagement, Engineering Judgement
7.4	The robot shall locate & classify at a distance > 75 mm.	Engineering Judgement
8	The robot shall locate & classify the Coloured Bins.	Stakeholder engagement
8.1	The Coloured Bins will be $200 \times 200 \times 200$ mm.	Stakeholder engagement
Object Manipulation		
9	The robot shall manipulate the Coloured Objects.	Stakeholder engagement
9.1	The shape of the Coloured Objects will be specified by Team 12.	Stakeholder engagement
9.2	The minimum size of the Coloured Objects to be manipulated shall be $20 \times 20 \times 20$ mm.	Collected Data
9.3	The robot shall manipulate at a height of 200mm above the floor.	Engineering Judgement
9.4	The maximum weight of the object to be manipulated shall be 150g.	Data sheet - myCobot 280pi
Continued on next page		

Table 1 – Continued from previous page

No.	REQUIREMENT	SOURCE
10	The robot shall transport the Coloured Objects.	Stakeholder engagement
10.1	The robot shall transport a minimum of 1 Coloured Object.	Stakeholder engagement
10.2	The robot should transport a maximum of 3 Coloured Objects.	Stakeholder engagement
11	The robot shall deposit the Coloured Objects into the corresponding Coloured Bins.	Stakeholder engagement
11.1	The coloured bin will have a 150mm square hole in the top.	Collected Data
Safety		
12	The robot shall be capable of being manually handled.	Risk Assessment
12.1	The robot will have a mass <20kg.	Risk Assessment
12.2	The robot should have a form factor that allows proper lifting technique.	Risk Assessment
12.3	The robot should not have any payloads which could cause injury.	Risk Assessment
13	The robot shall be able to be stopped in the case of unexpected behaviour.	Risk Assessment
13.1	The robot shall be capable of being issued a stop command wirelessly.	Risk Assessment
13.2	The robot shall have a form of easily accessible emergency shutdown button or switch.	Risk Assessment
Power		
14	The robot should operate without trailing cables.	Stakeholder Engagement
14.1	Onboard power should provide for a minimum of 25 minutes.	Engineering Judgement
14.2	Electrical design and maintenance shall comply with lab risk assessment.	Stakeholder Engagement

4 Requirements Verification Matrix

Table 2. Requirements Verification Matrix

Req #	Shall Statement	Verification Method	Verification of Success Criteria	Responsible Party	Results	Grade
Operation						
1	The robot shall operate within the Test Environment.	Demonstration	Robot operation is not impeded by factors of the testing environment.	All	TBC	Pass/Marginal/- Fail
2	The robot shall be capable of restarting the operation.	Demonstration	Robot can be shown to restart the operation after interruption/turn off state.	AB, QS	TBC	Pass/Marginal/- Fail
2.1	The robot shall achieve full operation in <5 minutes.	Test	During testing the robot can be restarted and fully operational within the 5 minute window.	AB, ZP	TBC	Pass/Marginal/- Fail
3	All hardware shall be mounted on the robot platform.	Demonstration	There is no un-mounted hardware on the robot.	AB, ZP	TBC	Pass/Marginal/- Fail
Navigation and planning						
4	The robot shall navigate autonomously.	Test	The robot successfully navigates the environment without external influence.	SH	TBC	Pass/Marginal/- Fail
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Table 2 – Continued from previous page

Req #	Shall Statement	Verification Method	Verification of Success Criteria	Responsible Party	Results	Grade
4.1	The robot shall avoid obstacles and unintended collisions.	Demonstration	During operation the robot does not impact the test environment wall, obstacles, coloured bins or boxes unintentionally.	SH, AB	TBC	Pass/Marginal/- Fail
4.2	The robot shall be capable of receiving commands from external controls.	Test	The robot responds accordingly to commands issued from external controls.	SH, QS	TBC	Pass/Marginal/- Fail
4.4	Maximum linear velocity shall be limited to ≤ 250 mm/s.	Test	The robot does not exceed 250 mm/s.	SH, AB	TBC	Pass/Marginal/- Fail
4.5	Maximum angular velocity shall be $<60^\circ/\text{s}$.	Test	The robot does not exceed $60^\circ/\text{s}$.	SH, AB	TBC	Pass/Marginal/- Fail
5	The robot shall locally generate paths for navigation.	Demonstration	The process of path planning is shown to be running locally.	SH, AB	TBC	Pass/Marginal/- Fail
5.1	Path planning shall use data from sensor data not pre-programmed global map.	Demonstration	Sensor data is shown to be driving decision making of the path planning.	SH, AB	TBC	Pass/Marginal/- Fail

Vision and perception

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Table 2 – Continued from previous page

Req #	Shall Statement	Verification Method	Verification of Success Criteria	Responsible Party	Results	Grade
6	The robot shall locate the Coloured Objects.	Test	The robot correctly locates the coloured objects within 3D space.	QS, AB	TBC	Pass/Marginal/- Fail
6.1	The robot shall determine the 3d coordinates (x,y,z) of the target relative to the camera.	Test	The robot accurately (within a grasp capable margin) provides the relative XYZ coordinates relative to the camera.	QS, AB	TBC	Pass/Marginal/- Fail
6.2	The robot shall locate coloured blocks of size > 20 × 20 × 20mm.	Demonstration	Coloured blocks of size 20 × 20 × 20mm shall be used in the coloured object testing to measure success.	QS, AB	TBC	Pass/Marginal/- Fail
7	The robot shall classify the Coloured Objects.	Test	The robot correctly classifies the coloured objects.	QS, AB	TBC	Pass/Marginal/- Fail
7.1	The robot shall classify coloured blocks of size > 20 × 20 × 20mm.	Demonstration	Coloured blocks of size 20 × 20 × 20mm shall be used in the coloured object testing to measure success.	QS, AB	TBC	Pass/Marginal/- Fail

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Table 2 – Continued from previous page

Req #	Shall Statement	Verification Method	Verification of Success Criteria	Responsible Party	Results	Grade
7.2	The robot perception shall tolerate lighting variation including shadows and differing shades of the same nominal colour.	Test	The robot can be shown to correctly identify the bins colour under the natural lighting conditions in the lab.	QS, AB	TBC	Pass/Marginal/- Fail
7.3	The robot shall locate & classify at a distance >75mm.	Demonstration	The robot provides accurate positional data and correctly identifies the objects colour whilst positioned adjacent to the coloured objects mounting box.	QS, AB	TBC	Pass/Marginal/- Fail
8	The robot shall locate & classify the Coloured Bins.	Test	The robot provides accurate positional data and correctly identifies the bins colour during testing.	QS, AB	TBC	Pass/Marginal/- Fail

Object manipulation

9	The robot shall manipulate the Coloured Objects.	Test	The robot successfully manipulates the Coloured objects in the desired way.	NL, ZP	TBC	Pass/Marginal/- Fail
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Table 2 – Continued from previous page

Req #	Shall Statement	Verification Method	Verification of Success Criteria	Responsible Party	Results	Grade
9.2	The minimum size of the Coloured Objects to be manipulated shall be 20 × 20 × 20mm.	Test	The robot successfully manipulates objects of the size specified.	NL, ZP	TBC	Pass/Marginal/- Fail
9.3	The robot shall manipulate at a height of 200mm above the floor.	Demonstration	The robot is shown to manipulate objects on top of the coloured block plinth and at the same height as the coloured deposit bins.	NL, ZP	TBC	Pass/Marginal/- Fail
9.4	The maximum weight shall be 150g.	Analysis	When calculated, none of the coloured blocks shall weigh more than 150g.	NL, ZP	TBC	Pass/Marginal/- Fail
10	The robot shall transport the Coloured Objects.	Test	The robot will successfully carry coloured objects between given locations.	NL, ZP, SH	TBC	Pass/Marginal/- Fail
10.1	The robot shall transport a minimum of 1 Coloured Object.	Test	The robot will show it can transport a single coloured object inside the test environment.	NL, ZP, SH	TBC	Pass/Marginal/- Fail

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Table 2 – Continued from previous page

Req #	Shall Statement	Verification Method	Verification of Success Criteria	Responsible Party	Results	Grade
11	The robot shall deposit the Coloured Objects into the corresponding Coloured Bins.	Test	For 3 successive trials the robot will deposit the Coloured object into the correct bin.	NL, ZP	TBC	Pass/Marginal/- Fail
Safety						
12	The robot shall be capable of being manually handled.	Demonstration	The robot shall be shown to be safely picked up and placed into the test environment.	NL, ZP	TBC	Pass/Marginal/- Fail
12.1	The combined weight of the robot and the payload shall be <20kg.	Analysis	When calculated the sum of weights of all the components and the payload shall be less than 20kg.	NL, AB	TBC	Pass/Marginal/- Fail
13	The robot shall be able to be stopped in the case of unexpected behaviour.	Demonstration	Emergency shutdown button demonstrated.	ZP	TBC	Pass/Marginal/- Fail
13.1	The robot shall be capable of being issued a stop command wirelessly.	Demonstration	The robot will show the ability to be stopped remotely.	SH, QS	TBC	Pass/Marginal/- Fail
13.2	The robot shall have a form of easily accessible emergency shutdown button or switch.	Inspection	Emergency stop button is present, indicated, and accessible.	ZP, SH	TBC	Pass/Marginal/- Fail

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Table 2 – Continued from previous page

Req #	Shall Statement	Verification Method	Verification of Success Criteria	Responsible Party	Results	Grade
Power						
14	Electrical design and maintenance shall comply with lab risk assessment.	Demonstration	The power supply and batteries can be shown to comply with the risk assessment by design and when inspected.	AB, ZP	TBC	Pass/Marginal/- Fail